



Full length article

FS-Diff: Semantic guidance and clarity-aware simultaneous multimodal image fusion and super-resolution

Yuchan Jie ^a, Yushen Xu ^b, Xiaosong Li ^b, Fuqiang Zhou ^c, Jianming Lv ^a, Huafeng Li ^d

^a School of Computer Science and Engineering, South China University of Technology, 510006, Guangzhou, China

^b School of Physics and Optoelectronic Engineering, Foshan University, 528225, Foshan, China

^c School of Instrumentation Science and Optoelectronics Engineering, Beihang University, 100191, Beijing, China

^d School of Information Engineering and Automation, Kunming University of Science and Technology, 650500, Kunming, China

ARTICLE INFO

Keywords:

Image fusion

Image super-resolution

Diffusion model

Vision language model

Benchmark

ABSTRACT

As an influential information fusion and low-level vision technique, image fusion integrates complementary information from source images to yield an informative fused image. A few attempts have been made in recent years to jointly realize image fusion and super-resolution. However, in real-world applications such as military reconnaissance and long-range detection missions, the target and background structures in multimodal images are easily corrupted, with low resolution and weak semantic information, which leads to suboptimal results in current fusion techniques. In response, we propose FS-Diff, a semantic guidance and clarity-aware joint image fusion and super-resolution method. FS-Diff unifies image fusion and super-resolution as a conditional generation problem. It leverages semantic guidance from the proposed clarity sensing mechanism for adaptive low-resolution perception and cross-modal feature extraction. Specifically, we initialize the desired fused result as pure Gaussian noise and introduce the bidirectional feature Mamba to extract the global features of the multimodal images. Moreover, utilizing the source images and semantics as conditions, we implement a random iterative denoising process via a modified U-Net network. This network is trained for denoising at multiple noise levels to produce high-resolution fusion results with cross-modal features and abundant semantic information. We also construct a powerful aerial view multiscale (AVMS) benchmark covering 600 pairs of images. Extensive joint image fusion and super-resolution experiments on six public and our AVMS datasets demonstrated that FS-Diff outperforms the state-of-the-art methods at multiple magnifications and can recover richer details and semantics in the fused images. The code is available at <https://github.com/XylonXu01/FS-Diff>.

1. Introduction

Owing to the limitations of optical imaging systems, images obtained via a single imaging device can only characterize part of a scene. For example, infrared (IR) images are limited by their specific spectral range and imaging resolution. They can present only some of the thermal radiation characteristics of a target and cannot reflect the texture and details of the visible image (VI). Multimodal image fusion (MMIF) is an essential field within image processing. It focuses on integrating complementary information from diverse sensors to generate a fusion result with a more complete representation of the scene [1–6]. Fused images with superior visual perception and excellent scene and target representation are widely used in fields such as biometrics [7], semantic segmentation [8], and target detection [9]. Medical image fusion

(MIF) and visible and infrared image fusion (VIRF) are challenging MMIF tasks.

MIF can simultaneously represent the metabolic and structural information in functional and anatomical medical images. By contrast, VIRF aims to preserve the high-spatial-resolution texture information in VI and the thermal target information in IR, as IR is not sensitive to light. In the past few decades, various MMIF techniques have been developed to enhance the quality of fused images. These fusion approaches can be classified as multiscale transform (MST)-based [10], sparse representation (SR)-based [11], saliency detection (SD)-based [12], and deep learning (DL)-based [13].

In contrast to the first three conventional fusion methods, DL-based methods can learn meaningful features from extensive training data

* Corresponding author.

E-mail addresses: jyc981214@163.com (Y. Jie), 2112355010@stu.fosu.edu.cn (Y. Xu), lixiaosong@buaa.edu.cn (X. Li), zfq@buaa.edu.cn (F. Zhou), jmlv@scut.edu.cn (J. Lv), hfcina99@163.com (H. Li).

¹ Equal Contribution

<https://doi.org/10.1016/j.inffus.2025.103146>

Received 26 December 2024; Received in revised form 19 March 2025; Accepted 20 March 2025

Available online 2 April 2025

1566-2535/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

to meet the requirements of sensitivity and fast adaptation of fusion methods in complex scenes [14]. Although existing MMIF can achieve satisfactory fusion outcomes in general, the following two challenges are neglected in practical applications. First, existing fusion methods suffer from poor performance in adaptively processing low-resolution (LR) source images, making it difficult to fulfill the requirements of applications such as unmanned aerial vehicle (UAV) reconnaissance, crop monitoring, and autonomous driving. In these applications, the state of the target or camera changes rapidly, increasing the demand for image information and resolution. In particular, when UAVs shootcapture images at high altitudes, multiple factors, such as flight altitude, steadiness, and atmospheric conditions, significantly affect image clarity. Therefore, in unfavorable imaging environments, such as long-distance shooting, and under the performance limitations of imaging systems, it is crucial to investigate methods that can adaptively perceive resolution degradation and flexibly incorporate semantic information while preserving cross-modal features. This is a challenge that urgently needs to be solved.

Generally, image fusion and super-resolution tasks can only be completed in steps. However, this approach can amplify errors from previous steps in subsequent steps, leading to a serious loss of modal information. Despite efforts to solve this problem [15–18], several challenges still remain. Xiao et al. [16] proposed a heterogeneous knowledge distillation network model. It attempts to obtain high-resolution (HR) fused images by using a teacher network to direct the student network through a corner-embedded attention mechanism. Li et al. [15] proposed a meta-learning-based fusion framework for fusing images with different resolutions. However, their generalizability and flexibility are limited on datasets of different types, failing to meet the high-magnification requirements of real scenes. When processing LR images with severely degraded resolution, these methods [15–17] struggle to adaptively perceive resolution and incorporate semantic information into the fused results. They also lack robust global and cross-modal feature extraction capabilities, making the fusion results susceptible to artifacts and hindering the recovery of high-fidelity details. This is mainly because high-magnification joint image fusion and super-resolution require not only pixel value adjustment but also full consideration of image semantics, details, and textures.

Second, despite advancements in specialized applications, existing aerial photography datasets present inherent limitations that hinder their direct adaptation to the MMIF task. For example, DroneVehicle [19] provides RGB-IR image pairs in diverse lighting conditions but lacks object diversity. Anti-UAV410 [20] only provides thermal infrared unimodal data and lacks visible counterparts. VEDAI [21] is limited by confined lighting conditions and a narrow range of target categories. Furthermore, although existing MMIF datasets including TNO [22], RoadScene [23], LLVIP [24], M³FD [25], and MSRS [26] are widely used, they are not specifically constructed for the dual tasks of simultaneous image super-resolution and fusion in long-distance shooting (e.g., aerial photography using a UAV). Specifically, these datasets lack coverage of degraded-resolution aerial images, diverse shooting distances, varied viewpoints, and a wide range of scenes. Additionally, MMIF has the potential to address the inconsistent performance of unimodal images in detection and segmentation tasks [25]. However, owing to the lack of annotated targets in available datasets, assessing the effectiveness of fusion approaches based on the performance of fused images in target detection or semantic segmentation tasks is difficult for the researchers.

Generative models are widely used in DL-based MMIF owing to their superior visual perception. These models effectively analyze the empirical distributions of complex scenes through an in-depth examination of the intrinsic laws and statistical properties of the images. Among these, generative adversarial networks (GANs)-based image fusion models [25,27,28] are predominant. Such models typically consist of a generator that is dedicated to fusion image generation and a discriminator that evaluates the distribution similarity between the

fused result and the source image. Despite the potential of GAN-based methods for generating satisfying fusion results, they experience unstable optimization, and their model-building process requires careful regularization and optimization strategies. In addition, GAN-based methods often face issues such as mode collapse and training instability [29]. As a successor to the GAN model, the denoising diffusion probabilistic model (DDPM) [30] recovers superior images via diffusion and denoising processes. Compared with GAN, DDPM exhibits a steadier optimization process and stronger interpretability. Zhao et al. [2] designed the fusion method utilizing DDPM, which splits the conditional generation problem into a maximum likelihood subproblem and an unconditional generation subproblem. This provides a new idea for the design of fusion methods based on generative models. Beyond these advancements, integrating generative models with vision-language model (VLM) offers new opportunities to enhance the quality of generated images. VLM can more precisely understand and generate content that aligns with specific semantics. By combining the strengths of the generative model with VLM, the field of MMIF can be advanced further.

To this end, we propose a novel approach called FS-Diff. It effectively utilizes generative priors and semantics to guide the model for clarity adaptive perception and intra- and inter-modal feature extraction to jointly achieve MMIF and super-resolution. FS-Diff is adaptive for sensing the degradation of resolution, with multimodal learning capabilities, and includes forward and backward Markovian diffusion processes. The forward Markovian process gradually adds Gaussian noise to the HR fusion image obtained by fusing the HR source images using CDD [31]. This process efficiently processes multimodal information, captures deep correlations between modalities, and achieves good fusion results. The reverse Markovian process connects the noisy target image of the forward process with the LR source image. It constructs a joint representation of the multimodal data in the latent space by introducing a bidirectional feature Mamba (BFM) block [32], which can perceive global information to train a modified U-Net network [33] to predict and eliminate the different levels of noise added by the forward process to recover the original HR fusion image. The inference phase starts with pure Gaussian noise. The HR fusion results are obtained by iteratively refining the noise removal by using a denoising network with multiple noise removal capabilities, conditioned on multimodal LR images and semantic guidance. To achieve precise semantic embedding during fusion, we aggregate semantic extraction and clarity perception capabilities through the proposed clarity sensing (CLSE) mechanism and the clarity-aware contrastive language-image pre-training (CA-CLIP) model. Through the cross attention module, deep semantics are effectively injected into the fusion features. The combination of the generative model with BFM and CLSE significantly enhances the ability of FS-Diff to handle cross-modal and detail feature extraction in the presence of image resolution degradation. As shown in Fig. 1, FS-Diff has high joint fusion and super-resolution performance with good generalization on VIRF, MIF, and multifocus tasks.

The main contributions of this paper are as follows.

1. We propose an end-to-end semantic-guided and resolution-aware fusion network. To the best of our knowledge, this is the first study to achieve synchronous fusion and super-resolution of pairs of LR multimodal image pairs with inconsistent clarity at multiple high magnifications.
2. We introduce the CLSE mechanism, where the core component is our proposed CA-CLIP, which is endowed with a clarity-adaptive sensing capability to extract semantic information from source images in advance to guide the fusion process.
3. We introduce the BFM to create a consolidated joint image representation during the fusion process to model and extract global features from multimodal images.

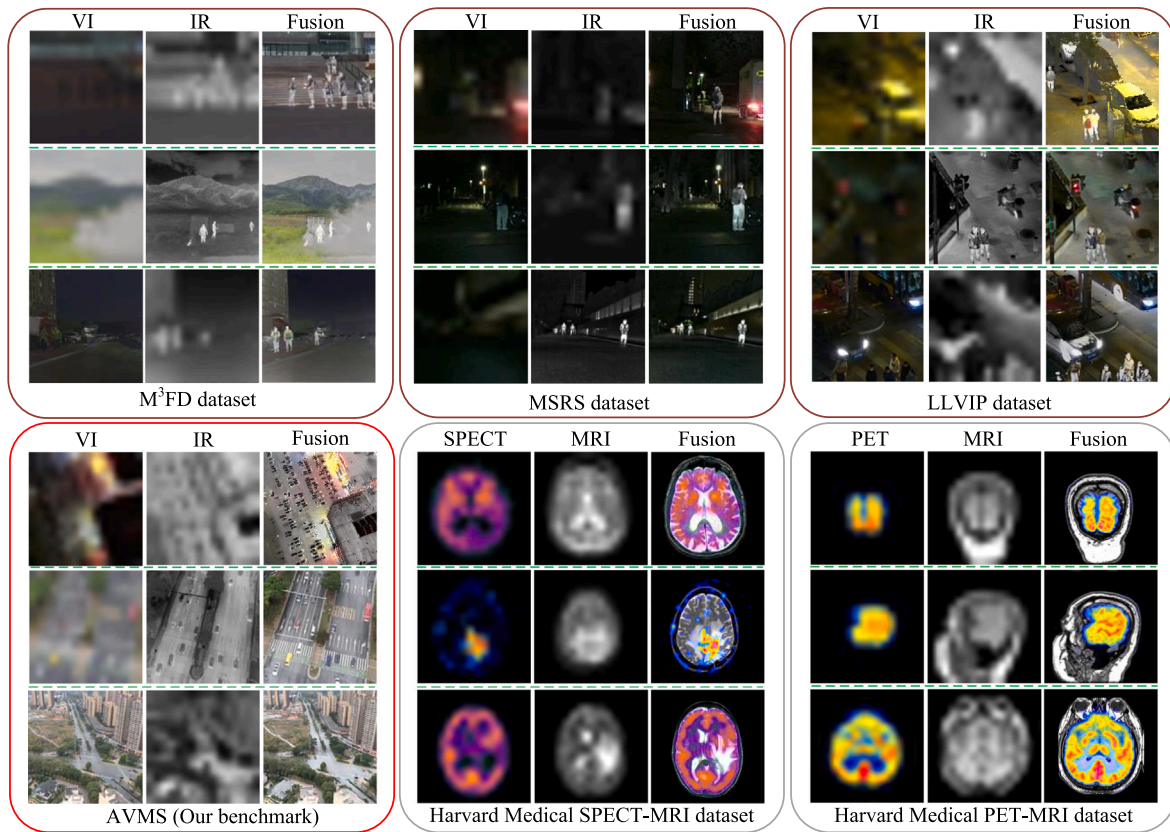


Fig. 1. Our FS-Diff demonstrates the outstanding fusion performance of joint image fusion and super-resolution with a scaling factor of 8 in five public and our AVMS datasets. The first four boxes show the VI-IR dataset, demonstrating three fusion scenarios: both VI and IR blurred, VI blurred, and IR blurred. The remaining two boxes present the multimodal medical dataset, where both images are blurred.

4. We construct an aerial view multiscene (AVMS) dataset with 600 aligned VI and IR image pairs, 8991 annotated targets that cover daytime, nighttime, dusk, and complex weather conditions, and more than seven different scenes. Our dataset is suitable for image fusion, super-resolution, long-distance detection, and semantic segmentation tasks.

The remainder of this paper is organized as follows. Section 2 presents a review of the research background. Section 3 details the motivation and algorithmic specifics of FS-Diff. Section 4 describes the created AVMS dataset. Section 5 presents the experimental validations. Section 6 compares the computational complexity of FS-Diff with those of other methods. Section 7 concludes the paper by outlining the contributions and application prospects.

2. Related work

2.1. Image fusion methods

2.1.1. Traditional fusion methods

Traditional image fusion methods include MST-based, SR-based, and SD-based approaches. Among these, MST-based methods [10,11] include three steps: decomposition, fusion, and reconstruction. First, typical MST tools, such as non-subsampled contourlet transforms [34] and edge-preserving filters [35,36], are introduced to decompose the image into multiscale components. Subsequently, specific fusion rules are used to measure the pixel activity of the image, and pixels or regions with high activity are selected. Finally, the fused image is reconstructed using the inverse MST. MST-based methods can reduce the generation of artifacts during the fusion process owing to translation invariance. However, inappropriate selection of the decomposition tool or number

of decomposition layers will lead to a loss of information in the fused image [14].

SR achieves an efficient and accurate representation of the image by decomposing it into a linear combination of sparse coefficients and an overcomplete dictionary. SR-based methods [37–39] assume that the activity level of an image can be computed in a sparse domain [40]. They first segment the image into multiple overlapping blocks via a sliding window, vectorize them. Then, they compute the sparse coefficients using sparse coding, and finally fuse them to reconstruct the fused image. Although SR-based methods have better noise robustness, balancing fusion performance and efficiency remains a challenge [41].

SD mimics the visual mechanism to highlight regions of an image that are significant to the surroundings [42]. It is widely used in tasks such as image compression [43] and segmentation [44]. SD-based methods [12] use saliency extraction operators to identify the salient regions of a source image, which in turn generates a decision map to reconstruct the result. These methods can be further classified into two categories namely, salient object detection [45–47] and salient edge detection [12,48], both of which consider the correlation between pixels in a local region. However, SD-based methods are sensitive to significance-detection algorithms and poor significance-extraction operators can lead to discontinuities at the regional junctions of the fused image.

2.1.2. Deep learning-based methods

In recent years, DL-based methods aimed at enhancing the perception of fused images, such as contrast and color, have emerged [15,48–58]. These methods can learn more effective fusion strategies under the constraint of a loss function owing to their powerful nonlinear fitting performance and learning abilities [59]. DL-based methods primarily

include autoencoder(AE) model-based [31,60–63], convolutional neural network (CNN)-based [64,65], and generation model (GEM)-based methods [66–70].

The AE-based method compresses the input data into a low-dimensional feature representation through an encoder and designs fusion layer rules to extract the features. These features are then remapped back into the data space by a decoder to complete image reconstruction. DenseFuse [71] is a typical AE-based method that trains a self-encoder using the MS-COCO dataset while combining addition and L1 norm strategies to obtain fused images. CDD [31] uses a self-coding network with a two-branch transformer-CNN structure and combines it with feature decomposition loss to extract cross-modal information. CNN-based methods utilize a well-designed network structure with loss functions to complete image feature extraction, fusion, and reconstruction. PMGI [72], as a representative method, divides the network into gradient and intensity paths to maintain the ratio of texture to intensity information in the fused image, and it reuses the features in the same path. To further improve the interpretability of the algorithm, LRR-Net [64] achieves fusion by mathematically modeling and analyzing the fusion task and establishing a link between the optimal solution and the network that implements it. CoCoNet [73] employs a coupled contrastive constraint loss and multilevel attention modules to preserve hierarchical features of different modalities, mitigating artifacts in the fused results. Text-IF [74] leverages semantic text guidance to achieve degradation-aware and interactive image fusion.

GEM-based methods, particularly GAN- and diffusion-model-based fusion methods, generate fused images that approximate the source image distribution through adversarial training and probabilistic modeling strategies, respectively. Typical methods, such as TCGAN [75] combine a conditional GAN and a transformer and design a wavelet fusion module to fuse cross-domain features. DDFM [2] decomposes the fusion task into two subproblems, namely, maximum likelihood and unconditional generation based on Bayesian modeling. It obtains the fused image using a natural image generation prior.

In addition, to meet the demands of advanced visual tasks, scholars have explored task-oriented image fusion methods. These methods apply high-level model semantic constraints to guide the training of fusion networks [25,76–78]. To enhance fusion image quality and the accuracy of subsequent perception tasks, CAF [79] introduces a perception-driven loss function, formulating fusion and perception as a joint problem. MRFS [80] enhances VIRF and segmentation performance by coupling these tasks and leveraging the intrinsic consistency between visual and semantic information. DetFusion [76] guides image fusion using target information obtained from target-detection network learning. Tang et al. [77] proposed combining fusion and semantic segmentation modules and increasing the semantic information obtained by the fusion module through semantic loss. Both methods [76,77] guided the fusion process by using information on the target location and semantic categories to improve the quality of the fused images.

Although the aforementioned methods can achieve satisfactory fusion results, they still face significant challenges in practical applications, particularly when dealing with LR MMIF. To address this challenge, some scholars have conducted preliminary studies on practice-oriented image fusion [81]. Yin et al. [17] proposed the interpolation of LR source images before fusing them by using an SR-based approach. Subsequently, Li et al. [15] proposed the joint learning of low-rank, sparse, and transformation dictionaries as well as learning compensation dictionaries that can reduce the loss of structure in the fused image and increase the loss of information during image super-resolution. However, this dual-task step-by-step processing method [15,16] has high computational complexity. It may also amplify artifacts generated in the super-resolution stage during the fusion stage. Moreover, this method cannot handle image fusion with high scaling factors.

2.2. Text to image models

VLMs offer significant opportunities for improving tasks involving general text and visual representations [82]. These models typically consist of a text encoder and an image encoder, which leverage contrastive learning to capture the aligned multimodal features between text and images [83]. As a typical example of VLMs, CLIP [83] employs image and text encoders and contrastive loss to align image–text pairs. It demonstrates robust zero-shot learning capabilities in diverse downstream tasks because of its unsupervised training method and construction of large-scale datasets. To further enhance CLIP’s performance on image restoration tasks, DA-CLIP emerged. DA-CLIP [84] integrates VLMs with low-level visual tasks, focusing on controlling the pretrained CLIP to predict degradation types and accordingly restore high-quality image features. CLIP-KD [85] enhances the performance of student models in zero-shot classification and cross-modal retrieval tasks through feature mimicry and interactive contrastive learning between the student and teacher encoders. Additionally, Stable Diffusion [86] achieves text-controlled image generation through the integration of attention and text encoding. By leveraging text guidance, it enables the customization of image generation, facilitating interactive control of multimodal fusion.

2.3. Diffusion models

Diffusion models [30] have been widely used in image super-resolution [86,87], image editing [88,89] and image inpainting [90] owing to their powerful generative abilities. In addition, the latent features of the data learned by the diffusion model during training can be used for discriminative tasks, such as image segmentation [91,92] and classification [93]. The central goal is to learn to recover the structure of the training data disturbed by noise via forward and backward diffusion processes. The forward process gradually adds different levels of noise to the training data over multiple time steps, whereas the reverse process learns an iterative denoising process via U-Net to recover the original image from Gaussian noise. The inference phase begins with Gaussian noise and iteratively estimates and removes noise to obtain the target output [94]. Similar to Langevin dynamics, the conditional diffusion probabilistic model (C-DDPM) [95] is based on conditional inputs and transforms from a standard normal distribution to an empirical data distribution through a multistage refinement process. DiffiT [96] achieves efficient high-fidelity image generation by introducing a time-dependent multihead self-attention mechanism for fine-grained control of the denoising process. The application of diffusion models to image fusion is promising.

Diffusion models have been widely applied in MMIF. Dif-Fusion [97] pioneered the integration of diffusion models into VIRF tasks, enhancing the color fidelity of the fused images. To optimize the efficient mapping of the original resolution image features in MMIF, LFDT-Fusion [98] introduced a latent feature-guided diffusion model for image fusion. Diff-IF [99] presented a fusion method based on a diffusion model and incorporating fusion knowledge priors. By decomposing the diffusion model into a conditional diffusion component and a fusion knowledge prior module, it facilitates the generation of high-fidelity fused results. Additionally, DRMF [100] proposed a degradation-robust conditional diffusion model to mitigate the impact of image degradation on the MMIF. However, current image fusion methods based on diffusion models are ineffective under conditions of single or dual-source image resolution degradation and neglect the preservation of semantic information. Moreover, MMIF and super-resolution are deterministic image-generation processes, and the output of a well-designed model should maintain a high degree of similarity to the source images. Therefore, utilizing the powerful generative capabilities of DDPM combined with VLM is essential in the fusion process. This allows the model to adaptively recognize resolution degradation and generate semantically rich fused images, thereby enhancing the accuracy of advanced visual tasks.

2.4. Mamba

The Mamba [101] model has the potential for long-sequence modeling, state space models (SSMs)-based models are motivated by continuous systems. It maps a one-dimensional sequence $x(t) \in \mathbb{R} \rightarrow y(t) \in \mathbb{R}$ through hidden states $h(t) \in \mathbb{R}^N$. The representation of the continuous system is as follows:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the evolution parameter, $\mathbf{B} \in \mathbb{R}^{N \times 1}$ and $\mathbf{C} \in \mathbb{R}^{1 \times N}$ are projection parameters.

To implement these systems in deep learning, the continuous parameters are transformed into discrete forms using a discretization step. A timescale parameter Δ is introduced to compute discrete versions of $\mathbf{A}$ and $\mathbf{B}$, denoted as $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$, respectively. The Zero-Order Hold method is employed for transformation, defined as:

$$\bar{\mathbf{A}} = \exp(\Delta \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I}) \cdot \Delta \mathbf{B} \quad (2)$$

Using discrete parameters, the continuous equations are reformulated into the following discrete-time version:

$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \quad y_t = \mathbf{C}h_t \quad (3)$$

Finally, the models output results through global convolution:

$$\bar{\mathbf{K}} = (\bar{\mathbf{C}}\bar{\mathbf{B}}, \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}}, \dots, \bar{\mathbf{C}}\bar{\mathbf{A}}^{M-1}\bar{\mathbf{B}}), \quad y = \mathbf{x} * \bar{\mathbf{K}} \quad (4)$$

where $\bar{\mathbf{K}} \in \mathbb{R}^M$ represents the structured convolution kernel and M denotes the length of the input sequence $\mathbf{x}$.

3. Method

3.1. Motivation

Although existing approaches [5,12,31,56,73,97,102] generally focus on improving the network's ability to extract detailed features from multimodal images, the MMIF task still faces three prominent challenges. First, in UAV aerial photography and fast unmanned scenarios, the problems of image structure destruction and resolution degradation due to long distances are particularly prominent. LR images not only weaken the clarity of the background and target in the fused image but also restrict the performance of the fused image in subsequent segmentation and detection tasks, thus restricting the scope of practical applications of image fusion. Current fusion methods [15–17] for LR images are only effective at low magnifications and cannot adapt to the serious degradation of image resolution, making it difficult to meet the performance and flexibility requirements in real scenes. Second, owing to the inherent variations across modalities in multimodal images, an effective fusion method that preserves both global and semantic information is crucial. However, current fusion techniques fail to adequately address these challenges, resulting in feature loss and diminished accuracy in subsequent visual tasks. Third, dataset construction is key to implementing the MMIF method in applications. From the UAV perspective, changes in flight altitude lead to variable image field-of-view sizes, which, in turn, cause variability in target sizes. This complexity further exacerbates the challenges of visual tasks, such as image fusion, semantic segmentation, and target detection, under multiple lighting conditions. However, the current publicly available MMIF datasets are not only limited in number but are also unavailable from aerial viewpoints, which severely limits the possibility of unmanned aerial systems realizing multimodal heterogeneous data fusion functions in agriculture, military, and other applications.

If the aforementioned challenges are not fully considered and effectively addressed, the performance of fusion algorithms in real applications will degrade significantly. Therefore, we propose FS-Diff. The core of FS-Diff is the use of the LR input image and the prior semantics obtained by the presented CLSE with CA-CLIP to constrain the diffusion

process, which alleviates the problems of the unstable training and performance degradation of DL-based methods. Moreover, to enhance the global and cross-modal feature perception of the fusion algorithm for multimodal images, we introduce the BFM to construct a joint representation of multimodal images.

3.2. FS-Diff

Fig. 2 shows the main principle of the proposed FS-Diff. The aligned input images with different clarity conditions are denoted by $\mathbf{x} \in \mathbb{R}^{3HW}$ and $\mathbf{y} \in \mathbb{R}^{HW}$, respectively, and the HR noise-free fused image, e.g. ground truth from CDD [31] is denoted by $\mathbf{F}_0 \in \mathbb{R}^{3HW}$, with H and W representing the height and width of the image, respectively. FS-Diff produces the fused result $\mathbf{F}_{fused}$ via T refinement iterations from an image initiated with complete noise: $\mathbf{F}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. By leveraging the BFM [32] to construct a joint representation of the inputs $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{F}_t$. On the basis of the studied conditional distribution $p_\theta(\mathbf{F}_{t-1} | \mathbf{F}_t, \mathbf{x}, \mathbf{y})$, FS-Diff iteratively optimizes the output to achieve the sequence output $(\mathbf{F}_{T-1}, \mathbf{F}_{T-2}, \dots, \mathbf{F}_{fused})$, and it eventually recovers $\mathbf{F}_{fused} \sim p(\mathbf{F}_{fused} | \mathbf{x}, \mathbf{y})$.

During the iterative refinement process, the distribution of the intermediate images is defined by the forward diffusion process $q(\mathbf{F}_t | \mathbf{F}_{t-1})$, and noise is gradually introduced into each output image via a fixed Markov chain. The goal of FS-Diff is to remove noise based on $\mathbf{x}$, $\mathbf{y}$ and semantic guidance e^{cl} iteratively by using a reverse Markov chain to recover the $\mathbf{F}_{fused}$ with rich semantic information and cross-modal features.

3.2.1. Bidirectional feature Mamba

To enhance FS-Diff's ability to extract global and information from source images, we use the BFM [32] block. First, $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{F}_t$ are concatenated along the channel dimension to form $\mathbf{s}_t$, which is fed into the BFM. The structure of the BFM is shown in Fig. 3.

First, $\mathbf{s}_t \in \mathbb{R}^{H \times W \times C}$ is reshaped into patches $\mathbf{s}_p \in \mathbb{R}^{J \times (P^2 \cdot C)}$, where C , H , W , J , and P denote the number of channels, height, width, the number of image patches, and the size of image patches, respectively. Then, $\mathbf{s}_p$ are linearly projected to the vector, and positional embeddings $\mathbf{E}_p$ are added:

$$\mathbf{S}_0 = [\mathbf{s}_{cls}; \mathbf{s}_p^1 \mathbf{W}; \mathbf{s}_p^2 \mathbf{W}; \dots; \mathbf{s}_p^J \mathbf{W}] + \mathbf{E}_p \quad (5)$$

where $\mathbf{W} \in \mathbb{R}^{(P^2 \cdot C) \times D}$ is a learnable projection matrix, and $\mathbf{s}_p^j$ represents the j th patch of $\mathbf{s}$. The class token $\mathbf{s}_{cls}$ denotes the whole patch sequence.

Next, the token sequence $\mathbf{S}_{l-1}$ is sent to the Vision Mamba encoder proposed in [32] to yield $\mathbf{S}_l$:

$$\mathbf{S}_l = \text{BFM}(\mathbf{S}_{l-1}) + \mathbf{S}_{l-1} \quad (6)$$

The normalization layer first normalizes $\mathbf{S}_{l-1}$ and projects it to $\mathbf{v}$ and $\mathbf{u}$. Then, $\mathbf{v}$ is processed in both directions. The 1-D convolution is applied to $\mathbf{v}$ to obtain $\mathbf{s}'_o$. Then, it is linearly projected into $\mathbf{B}_o$, $\mathbf{C}_o$, and $\mathbf{A}_o$. Using $\mathbf{A}_o$, $\bar{\mathbf{A}}_o$ and $\bar{\mathbf{B}}_o$ are derived. The outputs in the forward and backward directions are subsequently generated via the SSM. These outputs are then gated by $\mathbf{u}$ and combined to produce the final output token sequence $\mathbf{S}_l$.

Finally, to provide a more concise mathematical characterization of the BFM's role, we introduce $\mathbf{Y}$ to denote the BFM and $\hat{\mathbf{s}}_t$ to represent the global feature representation obtained by the BFM:

$$\hat{\mathbf{s}}_t = \mathbf{Y}(\mathbf{s}_t), \quad \mathbf{s}_t = \mathbf{x} \oplus \mathbf{y} \oplus \mathbf{F}_t \quad (7)$$

where $\oplus$ represents the concatenation operation. The target image with noise is $\mathbf{F}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

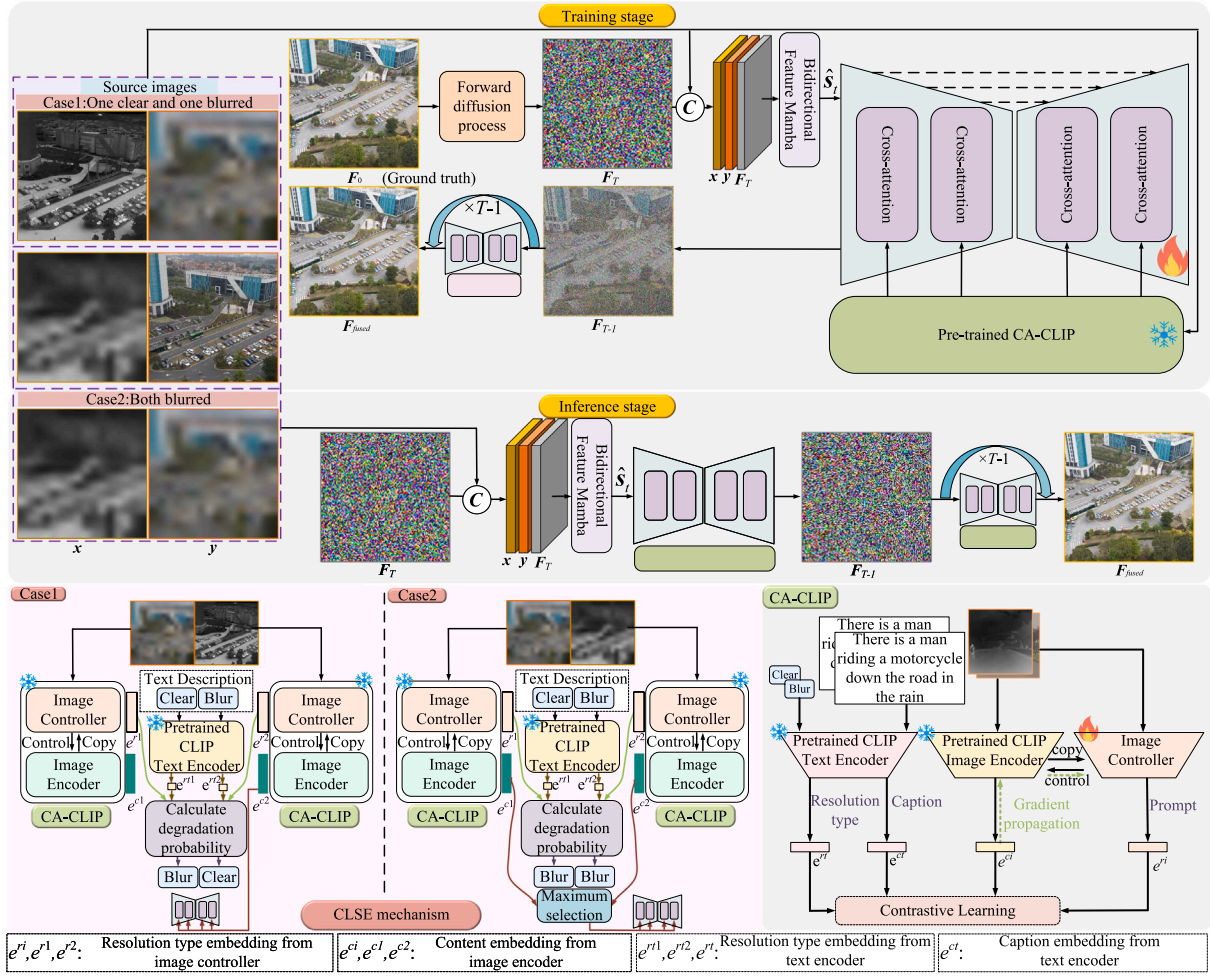


Fig. 2. The framework of FS-Diff. F_t and F_0 represent the noisy output image with t timesteps and the ground truth. The forward diffusion process $q(F_t | F_{t-1})$ adds Gaussian noise to F_0 , and the backward diffusion process $p_0(F_{t-1} | F_t, x, y)$ iteratively denoise. The CLSE mechanism dynamically selects semantic extraction strategies for multimodal images with inconsistent resolution. In Case 1 (single-image blur), CA-CLIP's image encoder injects the content embedding extracted from the clear image into the diffusion process. For Case 2 (dual-image blur), the model instead injects the maximum cross-image content embedding that is obtained from both blurred inputs.

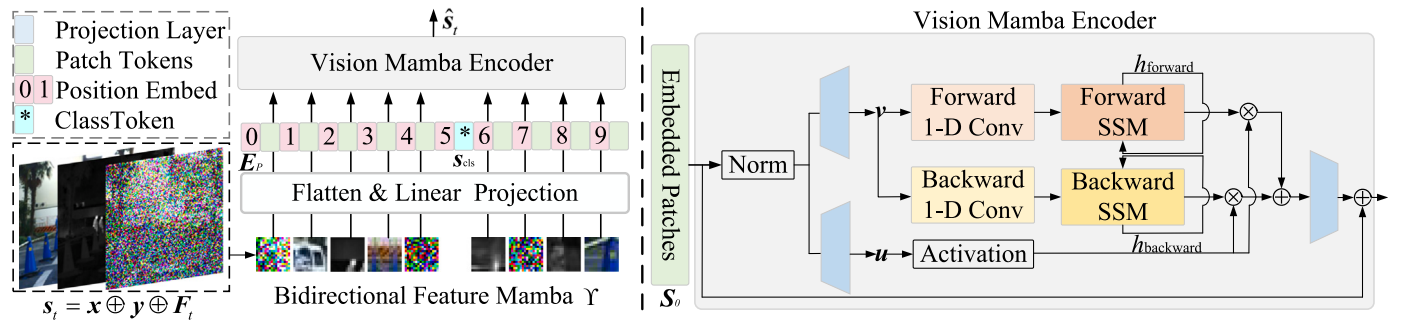


Fig. 3. The overview of the BFM.

3.2.2. CA-CLIP and the CLSE mechanism

This section introduces the CA-CLIP and the CLSE mechanism. By leveraging CLIP's [83] pre-trained cross-modal alignment capability, our framework enables precise semantic extraction from heterogeneous input modalities with varying clarity. The CLSE mechanism further enhances the model's interpretability and generalization capabilities by incorporating text description and pretrained CLIP text encoder. This design facilitates feature selection through semantic matching, enabling a deeper semantic-level analysis of the model's decision-making process. As a result, it not only improves model interpretability

but also demonstrates superior generalization performance in complex real-world scenarios.

Building upon DA-CLIP [84], CA-CLIP refines FS-Diff's semantic understanding of source inputs while regulating the generation process of HR fused results. Its core lies in leveraging CLIP's pretrained architecture to enable high-quality feature extraction from LR inputs. Simultaneously, it predicts the resolution type of the images (e.g., clear or blur), enabling the model to derive higher-quality semantic information from the clear image when only one of the images is blurred. As illustrated in Fig. 2, the structure of CA-CLIP involves the

matching of image content embedding e^{ci} and clear caption embedding e^{cl} . Furthermore, the resolution type embedding e^{ri} predicted by the image controller specifies the resolution type, which corresponds to the resolution type embedding e^{r1} from the text encoder. The image controller replicates the architecture of the image encoder, incorporating several connections initially set to zero to enhance its control capabilities and adjusting the outputs from each block of the encoder, thereby directing the encoder's predictive outputs [84].

In the process of simultaneous multimodal image fusion and super-resolution, effectively harnessing semantic information is crucial. The CLIP model [83], with its pretrained capability to align cross-modal features and extract rich semantic content, plays a pivotal role in computer vision. By integrating CLIP, our framework enables the precise extraction of semantic features from multimodal images. This is achieved even when the images exhibit inconsistent clarity, ultimately aiding in the generation of HR fused images.

In our proposed CLSE mechanism, CA-CLIP is first pretrained to acquire clarity-aware capabilities. Given text descriptions, e.g., “clear” and “blur”, the corresponding text features e^{r1} and e^{r2} are extracted by the pretrained CLIP text encoder. These features are then compared with the image features e^{r1} and e^{r2} , extracted by the image controller, to assess the semantic similarity between the image and the given text, thereby determining which modality corresponds to the clear image. In Case 1, to increase the semantic accuracy of the fused image and reduce the potential errors introduced by the blurred image, the feature of the clear image is extracted through the image encoder and injected into the diffusion process. In Case 2, where both images exhibit severe loss of semantic information with only luminance details preserved, the most prominent of the image features obtained from the image encoder for the two images is selected and similarly injected into the diffusion process. In addition, the cross-attention mechanism incorporates semantic cues into the fusion and super-resolution process. Here, the feature $\hat{s}_t$ from the BFM serves as the query vector, whereas the semantic cue e^{ci} serves as both the key and value vectors. The similarity computation between the query and the key reveals regions with high similarity scores, which signify a strong correspondence between $\hat{s}_t$ and e^{ci} . The model then assigns higher attention weights to these regions, thereby improving the recovery of fine details.

3.2.3. Diffusion process

The forward Markovian diffusion process q gradually adds Gaussian noise to the ground truth F_0 over T iterations [30]. At time step t , the noisy output image F_t can be expressed as

$$q(F_{1:T} | F_0) = \prod_{t=1}^T q(F_t | F_{t-1}) \quad (8)$$

$$q(F_t | F_{t-1}) = \mathcal{N}(F_t | \sqrt{\alpha_t} F_{t-1}, (1 - \alpha_t) I) \quad (9)$$

where $\alpha_{1:T}$ ($0 < \alpha_t < 1$) represents the hyperparameters that affect the noise variance that needs to be added at each iteration, I is the unit matrix. F_t and F_{t-1} denote the outputs obtained at time steps t and $t - 1$, respectively.

Notably, the intermediate iterative process describing F_t can be marginalized for a given F_0 condition:

$$q(F_t | F_0) = \mathcal{N}(F_t | \sqrt{\gamma_t} F_0, (1 - \gamma_t) I) \quad (10)$$

where $\gamma_t = \prod_{i=1}^t \alpha_i$, γ_t is the product of all α_i from the initial time step to the current time step t . F_0 is the HR noise-free fused image.

3.2.4. Optimization of the denoising model

The neural denoising model f_θ supplies an approximation of the gradient of the logarithmic density of the data and plays a vital role in FS-Diff. We refine a neural denoising model f_θ to predict the noise ϵ , in each forward diffusion process and thus perceive the noise level.

Algorithm 1 Training

```

1: repeat
2:    $(\mathbf{x}, \mathbf{y}, F_0) \sim p(\mathbf{x}, \mathbf{y}, F)$ 
3:    $\gamma_t \sim p(\gamma)$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, I)$ 
5:   Compute the gradient by
6:    $\nabla_\theta \|f_\theta(\hat{s}_t, \gamma_t, e^{ci}) - \epsilon\|_2^2$ 
7: until converged

```

To enable FS-Diff to more flexibly incorporate semantic information, we integrate a cross-attention mechanism with U-Net of f_θ . The U-Net architecture consists of an encoder–decoder structure connected by a bottleneck. The encoder extracts high-level features, whereas the decoder progressively reconstructs them to the original resolution. Skip connections preserve multilevel features between the encoder and decoder. Noise-level embeddings are incorporated to handle varying noise levels. Semantic conditions are injected into the diffusion process via a cross-attention mechanism, enhancing the semantic consistency of the fused image.

In addition, $f_\theta(\hat{s}_t, \gamma_t, e^{ci})$ contains three inputs: the joint global representation $\hat{s}_t$, the noise variance γ_t and the image content embedding e^{ci} . Algorithm 1 shows the total training process, the Kaiming initialization method [103] is used to initialize the conditional noise predictor f_θ . The training process begins by sampling the LR image pairs $\mathbf{x}, \mathbf{y}$ and the initial HR target image F_0 from the joint distribution $p(\mathbf{x}, \mathbf{y}, F)$. The noise intensity γ_t is subsequently randomly sampled from the probability distribution $p(\gamma)$, followed by sampling noise ϵ from the standard normal distribution. The gradient is then computed based on the L_2 error between the predicted noise from f_θ and the true noise ϵ . This iterative process continues until convergence is achieved.

The object function of FS-Diff is denoted as follows:

$$L = \mathbb{E}_{Y(\mathbf{x}, \mathbf{y}, F_t), \epsilon, t} [\|f_\theta(\hat{s}_t, \gamma_t, e^{ci}) - \epsilon\|_2^2] \quad (11)$$

$$e^{ci} = \begin{cases} e^{c1}, & \text{if } C_{VI} \wedge B_{IR}, \\ e^{c2}, & \text{if } C_{IR} \wedge B_{VI}, \\ \max(e^{c1}, e^{c2}), & \text{if } B_{VI} \wedge B_{IR}, \end{cases} \quad (12)$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, I)$. It is added to the HR target image to generate a noisy version, and f_θ is trained to predict this noise. $\mathbb{E}$ represents the expected value. B_{VI} and B_{IR} denote blurred VI and IR, respectively, whereas C_{VI} and C_{IR} represent clear VI and IR, respectively. e^{c1} and e^{c2} represent the features of VI and IR, respectively, from the image encoder of a pretrained CA-CLIP. $\wedge$ denotes that the conditions are satisfied simultaneously.

3.2.5. Inference

Algorithm 2 The inference of FS-Diff

```

1:  $F_T \sim \mathcal{N}(\mathbf{0}, I)$ 
2: for  $t = T, T - 1, \dots, 1$  do
3:   if  $t > 1$  then
4:      $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$ 
5:   else
6:      $\mathbf{z} = \mathbf{0}$ 
7:   end if
8:    $F_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( F_t - \frac{1 - \alpha_t}{\sqrt{1 - \gamma_t}} f_\theta(\hat{s}_t, \gamma_t, e^{ci}) \right) + \sqrt{1 - \alpha_t} \mathbf{z}$ 
9: end for
10: return  $F_{fused}$ 

```

The inference process of FS-Diff is defined as a reverse Markovian process, as opposed to a forward diffusion process. Our goal was to model the distribution of the fused result F_{fused} for the given $\mathbf{x}$ and $\mathbf{y}$ conditions. That is, $p(F_{fused} | \mathbf{x}, \mathbf{y})$. The detailed computational

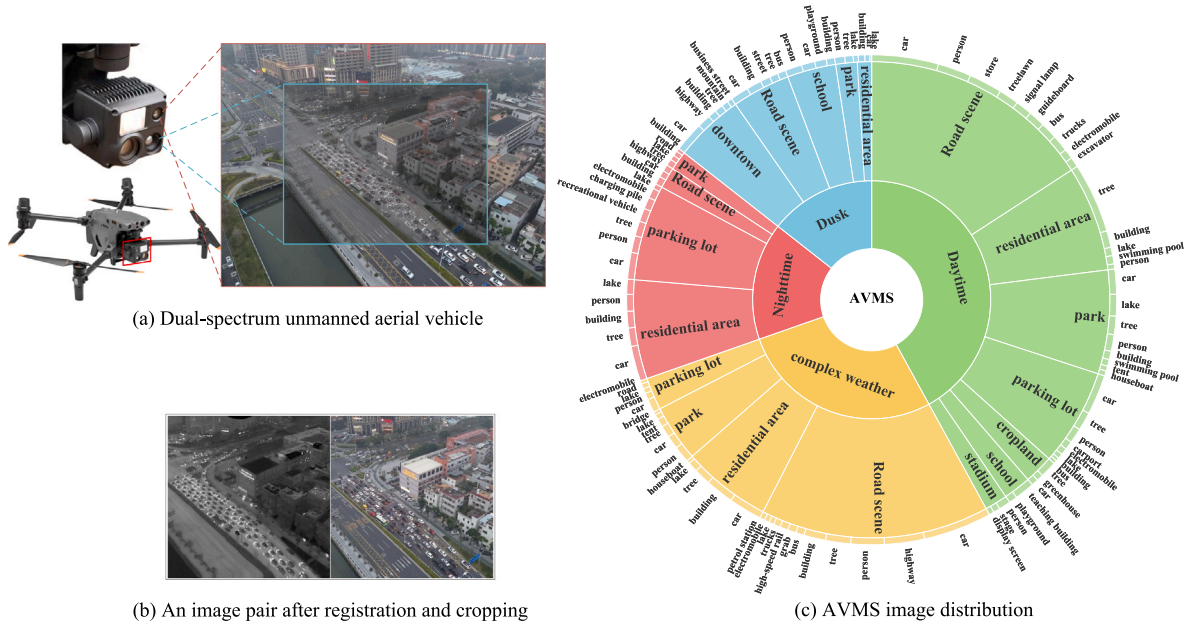


Fig. 4. (a) Image pair acquisition, (b) post-processing, and (c) scene distribution for AVMS.

procedure for the inference is described in Algorithm 2, which starts with the Gaussian noise F_T , where $p(F_T) = \mathcal{N}(F_T | \mathbf{0}, \mathbf{I})$. For each time step t (from T to 1), if $t > 1$, a random noise vector $\mathbf{z}$ is sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$. When $t=1$, $\mathbf{z}$ is set to 0. Same as in [30], the vector $\mathbf{z}$ represents the noise introduced during the inference process. ϵ is subsequently estimated via f_θ , and F_t is iteratively refined. Ultimately, this iterative process yields the final HR fused image, F_{fused} .

Each iteration generates output images $F_t \rightarrow F_{t-1}$ with different noise levels, and as t decreases, the noise level decreases. At the end of the iteration, F_{fused} is returned as the final clear fusion result, the mathematical expression is as follows:

$$p_\theta(F_{0:T} | \mathbf{x}, \mathbf{y}) = p(F_T) \prod_{t=1}^T p_\theta(F_{t-1} | F_t, \mathbf{x}, \mathbf{y}) \quad (13)$$

$$p_\theta(F_{t-1} | F_t, \mathbf{x}, \mathbf{y}) = \mathcal{N}(F_{t-1} | \mu_\theta(\hat{s}_t, \gamma_t, e^{ci}), \sigma_t^2 \mathbf{I}) \quad (14)$$

$$\sigma_t^2 = \frac{(1 - \gamma_{t-1})(1 - \alpha_t)}{1 - \gamma_t} \quad (15)$$

$$\mu_\theta(\hat{s}_t, \gamma_t, e^{ci}) = \frac{1}{\sqrt{\alpha_t}} \left(F_t - \frac{1 - \alpha_t}{\sqrt{1 - \gamma_t}} f_\theta(\hat{s}_t, \gamma_t, e^{ci}) \right) \quad (16)$$

where σ_t^2 and $\mu_\theta(\hat{s}_t, \gamma_t, e^{ci})$ represent the variance and the mean of $p_\theta(F_{t-1} | F_t, \mathbf{x}, \mathbf{y})$, respectively.

The $f_\theta(\cdot)$ is trained to predict ϵ . F_{fused} can be approximated under the given condition for F_t .

$$\hat{F}_{fused} = \frac{1}{\sqrt{\gamma_t}} \left(F_t - \sqrt{1 - \gamma_t} f_\theta(\hat{s}_t, \gamma_t, e^{ci}) \right) \quad (17)$$

where $\hat{F}_{fused}$ is the approximation of F_{fused} .

Finally, the refinement steps for each iteration are as follows:

$$F_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}} \left(F_t - \frac{1 - \alpha_t}{\sqrt{1 - \gamma_t}} f_\theta(\hat{s}_t, \gamma_t, e^{ci}) \right) + \sqrt{1 - \alpha_t} \epsilon_t \quad (18)$$

where $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

4. Multimodality image fusion benchmark

Multimodal datasets are crucial for building accurate and efficient deep learning-based MMIF systems [24]. However, most existing multimodal image datasets [22–26] mainly rely on handheld IR or VI

cameras, which have a relatively limited image field of view. In real-world complex environments, such as battlefield environments, UAVs need to adjust the shooting distance and camera parameters in real time to cope with changing terrain and weather, often resulting in the loss of semantic information and local structural features. This significantly affects the accuracy of advanced vision tasks. In recent years, several aerial datasets have been proposed for specific tasks, such as vehicle detection [19], drone tracking [20], and broader object detection tasks [21,104]. Anti-UAV410 [20] provides a thermal IR UAV tracking benchmark covering lakes, forests, and mountains, offering valuable support for anti-UAV tracking algorithm development. However, it is limited to IR images and does not support multimodal image fusion. VEDAI [21] focuses on small vehicle detection from an aerial perspective, including both large and small-sized IR and VI image pairs. Nonetheless, its limited illumination conditions and few nighttime scenes restrict its application in low-light scenarios. DVTOD [104] introduces a misaligned visible-thermal dataset with 54 scenes and 16 challenging attributes, but the misalignment adds complexity to the multimodal fusion algorithm design. DroneVehicle [19] provides a drone-based RGB-IR image pair for vehicle detection under varying lighting conditions from day to night, yet its narrow target category limits its use in broader tasks.

Although these datasets contribute significantly to object detection and tracking, they fail to support the comprehensive training and evaluation of multimodal fusion algorithms. To address these challenges, we established the AVMS dataset, which has a wide range of shooting angles, flexible shooting angles, and diverse lighting conditions. Unlike existing datasets, AVMS incorporates various scenes with multiple shooting distances, providing data support for developing practical deep-learning-based MMIF and super-resolution methods. AVMS is expected to be embedded in unmanned aerial systems with multisource heterogeneous data fusion functions in the future, which will provide key technical support for intelligent battlefields and UAV crop inspection.

A professional UAV was used for image acquisition, as shown in Fig. 4(a). The UAV can fly and shoot steadily in windy, snowy, rainy, and low-light conditions to obtain the correct target size, and the drone can be controlled at an altitude of 5–20 m. The standard resolutions of the captured images were 1280×1024 and 4000×3000 . To address the issue of the inconsistent field-of-view size of the built-in camera of the UAV, we cropped and registered the VI and IR images, as shown in Fig.

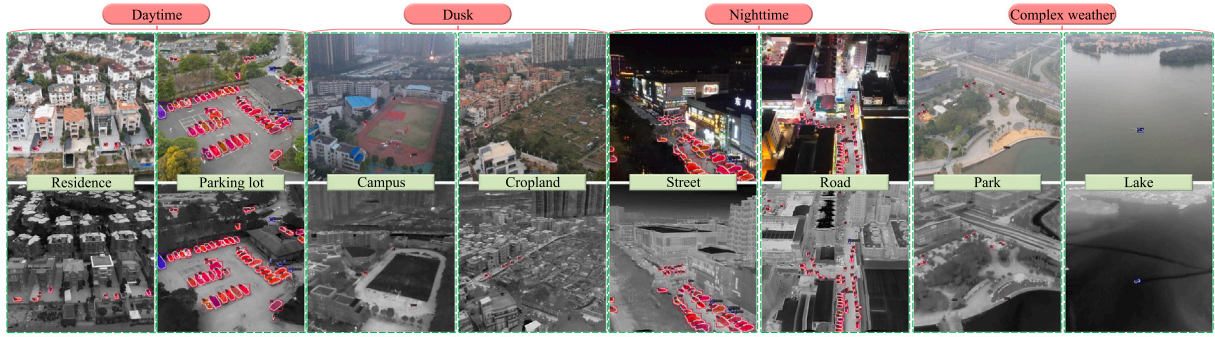


Fig. 5. Viewable distribution of scenes in our AVMS dataset. The first line is the visible images, and the second line is the corresponding infrared images. The text in the green box represents the scene type.

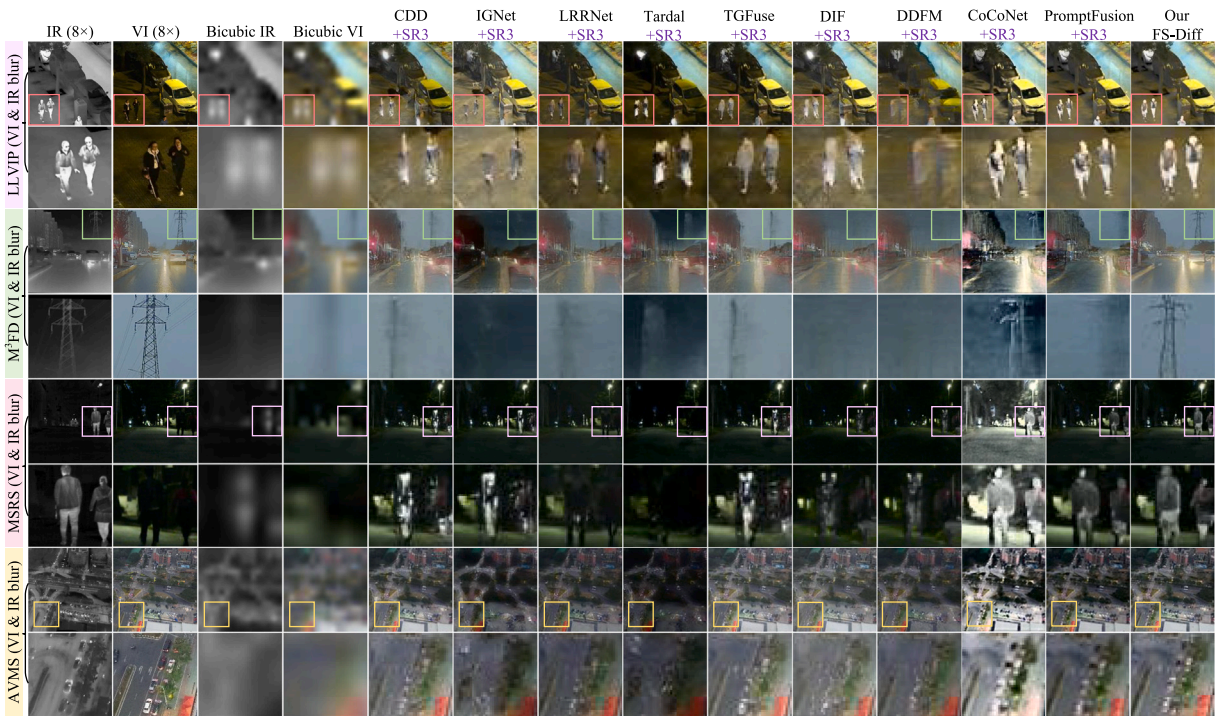


Fig. 6. Fusion and super-resolution results for different methods on LLVIP, M³FD, MSRS and our AVMS datasets (scale:8).

4(b). In addition, to obtain aligned image pairs with different clarity, the two modal images were downsampled via bicubic interpolation.

The 600 pairs of images contained in the AVMS dataset were classified into four categories on the basis of illumination levels: daytime, dusk, nighttime, and complex weather. Fig. 5 shows eight subscenes: residences, roads, campuses, parks, streets, stadiums, parking lots, croplands, and diverse environments in both urban and rural areas. Notably, MSIF aims to improve the accuracy of advanced vision tasks, such as detection and segmentation, to increase the effectiveness of applications in real-world scenarios, such as autonomous driving and drone combat. To this end, the AVMS dataset provides 8991 annotated targets, including 8046 cars, 113 buses, 157 trucks, 4 electric vehicles, 526 people, 138 tents, and 7 boats, laying a critical data foundation for subsequent target detection and semantic segmentation tasks.

5. Experiments

This section is organized as follows. Section 5.1 details the experimental setup. Sections 5.2 and 5.3 outline comparative experiments with state-of-the-art methods on the VIRF and MIF tasks, respectively,

and provide quantitative versus qualitative analyses. Section 5.4 explores the validity of the BFM block and CLSE mechanism via ablation experiments. Section 5.5 evaluates the performance of the model on an advanced visual task. Finally, Section 5.6 outlines extended FS-Diff experiments on multifocus image fusion.

5.1. Experimental setups

(1) **Datasets:** Our experiments were conducted using four VIRF datasets: LLVIP (training: 12025, test: 3463) [24], M³FD (training: 3780, test: 300) [25], MSRS (training: 1083, test: 362) [26], and our AVMS (training: 400, test: 200). MRI-PET (training: 287, test: 32) and MRI-SPECT (training: 497, test: 55) of the Harvard medical image dataset [105] were selected for the MIF experiments.

(2) **Image processing:** The HR fusion image was obtained via CDD [31], and the LR source image was down-sampled from the HR source image via a bicubic approach to synthesize the “LR multi-modal image-HR fusion image” training set. For the test set, similar to the configuration in [106–108], we cropped the three public datasets to 16×16 , degraded them bicubically, and named them LLVIP-val, M³FD-val, and MSRS-val.

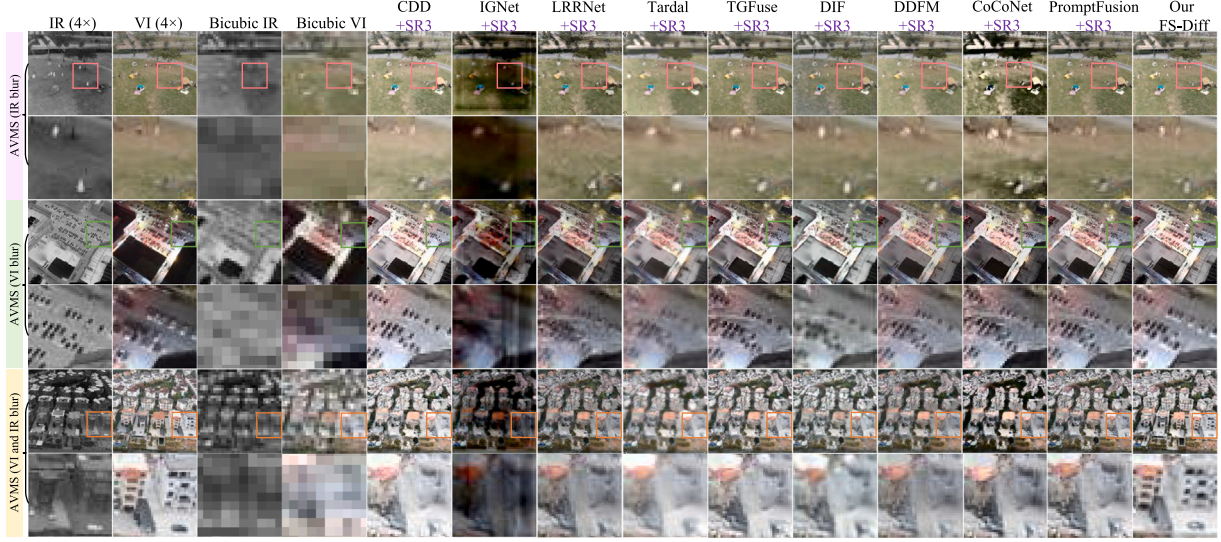


Fig. 7. Fusion and super-resolution results for different methods on our AVMS dataset (scale:4).

(3) Compared methods: We compared FS-Diff with 12 state-of-the-art (SOTA) image fusion approaches whose codes are publicly available in a VIRF task. We classified them into five groups. The first group is AE-based methods, and includes CDD [31], UMF [56] and PromptFusion [109]. The second group is CNN-based methods and consists of CoCoNet [73], ReConet [55], IGNet [110], and LRRNet [64]. The third group is GAN-based methods and includes Tardal [25] and TGFuse [111]. The fourth group is diffusion model-based methods and includes DIF [97] and DDFM [2]. The fifth group is MetaFusion [15], a SOTA joint image super-resolution and fusion method. Moreover, we compared FS-Diff with SOTA methods, including CDD [31], BDLFusion [50], U2Fusion [23], and DDFM [2] which are applicable to the MIF task, to prove its effectiveness in MIF.

(4) Metrics: To objectively and comprehensively evaluate the fused results, visual information fidelity (VIF) [112], $Q^{AB/F}$ [113], structural similarity index measure (SSIM) [114], peak signal to noise ratio (PSNR), learned perceptual image patch similarity (LPIPS) [115], and mean squared error (MSE) were used. The VIF, SSIM, and PSNR are image fidelity measures, whereas $Q^{AB/F}$ is evaluated by quantifying the edge information, LPIPS measures the perceived quality of the image, and MSE corresponds to the error in predicting the image.

(5) Implementation details: Our model was trained on four NVIDIA GeForce RTX 3090 GPUs and tested on a single RTX 3090 GPU. During training, we considered various scaling factors, including $\times 2$, $\times 4$, and $\times 8$, for a total of 800,000 training steps with a batch size of 80. The Adam optimizer was adopted with a fixed learning rate of 1×10^{-4} . Adapting the parameter configuration proposed in [116], we employ $T = 4000$ to maintain an optimal trade-off between fusion accuracy and inference speed.

With the exception of MetaFusion [15], other comparison methods cannot be directly applied to the joint task of fusion and super-resolution. Therefore, these methods perform super-resolution followed by fusion in a stepwise manner for comparison with FS-Diff. Among various image super-resolution methods [95,117], given the superior performance of SR3 across multiple scaling factors, we selected it as the baseline for super-resolution. Specifically, LR source image pairs were initially super-resolved to obtain HR images via SR3 [95], which was retrained on MMIF images, and then these HR images were fused to obtain the final fused image.

5.2. IR and VI image fusion

5.2.1. Subjective assessment

Figs. 6–8 show the fusion and super-resolution results of several SOTA methods with relatively better visual performance at the $\times 8$, $\times 4$, and $\times 2$ scales. As seen from the enlarged area, the compared methods exhibit a degree of blurriness and a deficiency in fine-grained details and fail to retain the structures of “person” in LLVIP and MSRS, “electrical pole” in M³FD and “cars” in AVMS.

The fusion results of IGNet+SR3, DIF+SR3, and DDFM+SR3 on the LLVIP dataset exhibit significant semantic errors in reconstructing “person” and “car”. Compared with the other methods, CoCoNet+SR3 and PromptFusion+SR3 exhibit good detail preservation of “person” in the LLVIP and MSRS datasets. Moreover, CDD+SR3, IGNet+SR3, Tardal+SR3, and CoCoNet+SR3 demonstrate artifacts in smooth areas, such as roadways in the M³FD dataset. Conversely, FS-Diff demonstrates greater stability in smooth areas and preserves richer contrast and significant information, highlighting its notable advantages in various fusion and super-resolution scenarios.

5.2.2. Objective assessment

Tables 1, 2, 3 and 4 show the VIF, $Q^{AB/F}$, SSIM, PSNR, LPIPS, and MSE values for VIRF and super-resolution on three public datasets and our constructed datasets, under three different resolution degradation scenarios of the source images. The following observations were made: (1) CDD+SR3 and PromptFusion+SR3 demonstrate strong performance in terms of VIF and PSNR, effectively preserving visual information fidelity. ReConet+SR3 and DDFM+SR3 outperform UMF+SR3 in terms of the MSE. On the LLVIP dataset, LRRNet+SR3 achieves commendable PSNR scores. In the context of blurred IR images at a $\times 4$ scaling factor, TGFuse+SR3 and PromptFusion+SR3 show robust overall performance. By contrast, DIF+SR3, IGNet+SR3, and Tardal+SR3 yield relatively average results. (2) As a SOTA joint image fusion and super-resolution method, MetaFusion is not capable of efficiently fusing images with severely degraded resolution, and performs poorly on six metrics. (3) FS-Diff performed well for the VIF, $Q^{AB/F}$, SSIM, and LPIPS metrics. Overall, FS-Diff, which integrates both image fusion and super-resolution tasks in a unified framework, demonstrates superior performance across all four datasets compared with methods that process these tasks separately. This fully demonstrates its ability



Fig. 8. Fusion and super-resolution results for different methods on our AVMS dataset (scale:2).

Table 1

The objective results for infrared and visible image fusion and super resolution ($\times 8$, Red: the best, Blue: the second, Green: the third).

Dataset: MSRS fusion dataset (VI and IR blurred scale:8)							Dataset: M ² FD fusion dataset (VI blurred scale: 8)						
Methods	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓	Methods	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
CDD+SR3	0.465	0.417	1.139	15.944	0.272	2555.251	CDD+SR3	0.303	0.363	0.946	13.173	0.296	3844.906
Reconet+SR3	0.335	0.315	0.561	15.221	0.320	2040.049	Reconet+SR3	0.251	0.293	0.968	12.037	0.337	3157.553
IGNet+SR3	0.364	0.319	0.885	16.644	0.281	2000.401	IGNet+SR3	0.463	0.436	0.996	12.146	0.261	4283.597
UMF+SR3	0.241	0.219	0.961	15.742	0.257	1614.521	UMF+SR3	0.442	0.388	1.135	13.617	0.233	2390.301
LRRNet+SR3	0.359	0.367	0.985	15.607	0.266	2399.006	LRRNet+SR3	0.187	0.266	0.874	13.161	0.299	3475.444
Tardal+SR3	0.129	0.133	0.191	14.008	0.337	3199.039	Tardal+SR3	0.272	0.272	0.494	10.108	0.435	6311.403
TGFuse+SR3	0.454	0.414	1.146	16.233	0.273	2454.667	TGFuse+SR3	0.279	0.394	0.965	13.017	0.279	3951.056
DDFM+SR3	0.397	0.356	1.136	17.407	0.258	1770.990	DDFM+SR3	0.425	0.391	1.041	14.162	0.256	2820.482
DIF+SR3	0.462	0.414	1.157	15.887	0.258	2620.435	DIF+SR3	0.298	0.384	0.977	12.469	0.297	4372.365
CoCoNet+SR3	0.361	0.267	0.612	9.612	0.384	7245.056	CoCoNet+SR3	0.335	0.284	0.704	11.690	0.374	4463.171
PromptFusion+SR3	0.465	0.415	1.163	16.238	0.267	2470.622	PromptFusion+SR3	0.295	0.347	0.936	13.143	0.299	3802.862
MetaFusion	0.102	0.055	1.014	13.667	0.457	2098.653	MetaFusion	0.362	0.350	1.084	11.939	0.312	3092.041
Our FS-Diff	0.486	0.481	1.165	15.858	0.262	2632.360	Our FS-Diff	0.563	0.531	1.271	12.828	0.212	4044.098
Dataset: AVMS fusion dataset (VI blurred scale: 8)							Dataset: LLVIP fusion dataset (IR blurred scale: 8)						
Methods	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓	Methods	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
CDD+SR3	0.518	0.493	1.303	15.173	0.236	2414.617	CDD+SR3	0.571	0.546	1.148	14.366	0.277	2511.818
Reconet+SR3	0.446	0.424	1.275	14.804	0.322	1818.363	Reconet+SR3	0.363	0.296	0.751	12.440	0.360	2284.514
IGNet+SR3	0.502	0.462	1.096	13.171	0.249	3183.666	IGNet+SR3	0.481	0.480	1.047	14.929	0.284	2114.455
UMF+SR3	0.507	0.457	1.377	15.809	0.211	1389.422	UMF+SR3	0.364	0.331	1.057	14.926	0.283	1498.730
LRRNet+SR3	0.408	0.400	1.140	14.938	0.252	2536.516	LRRNet+SR3	0.359	0.341	0.929	15.573	0.297	1838.926
Tardal+SR3	0.341	0.267	0.635	10.825	0.404	5132.788	Tardal+SR3	0.223	0.262	0.465	12.933	0.421	3225.795
TGFuse+SR3	0.507	0.546	1.300	15.098	0.230	2281.759	TGFuse+SR3	0.534	0.549	1.118	14.781	0.283	2077.500
DDFM+SR3	0.498	0.481	1.337	16.030	0.224	1906.830	DDFM+SR3	0.508	0.485	1.135	14.877	0.276	2437.723
DIF+SR3	0.484	0.516	1.270	14.629	0.252	2538.968	DIF+SR3	0.507	0.511	1.100	13.792	0.299	2579.120
CoCoNet+SR3	0.437	0.344	0.998	13.069	0.304	3216.136	CoCoNet+SR3	0.465	0.388	0.884	11.088	0.325	5370.172
PromptFusion+SR3	0.551	0.509	1.307	15.281	0.232	2480.831	PromptFusion+SR3	0.567	0.543	1.169	14.899	0.271	2256.018
MetaFusion	0.427	0.359	1.108	13.572	0.307	1897.778	MetaFusion	0.385	0.221	0.999	12.185	0.382	2217.690
Our FS-Diff	0.559	0.534	1.334	14.867	0.209	2389.134	Our FS-Diff	0.480	0.481	1.140	13.806	0.262	2488.696

to generate high-resolution fused images that are consistent with the visual perception of the human eye, with a more complete retention of details and semantic information.

Subjective and objective evaluations simultaneously affirmed the applicability of FS-Diff for image fusion and super-resolution at different magnifications in different scenes. This is primarily because FS-Diff is trained based on joint multitask learning, including fusion and super-resolution, which strengthens the feature recognition ability of the model.

5.3. Medical image fusion

The results of the subjective and objective assessments of the MIF task, under the condition where both image modalities are blurred, are presented in Table 5 and Fig. 9. In Fig. 9, the comparison methods are fuzzy or even lost when processing structure and color information. For example, BDLFusion, U2Fusion, and DDPM fail to effectively retain marginal skeletal information on MRI, whereas CDD exhibits poor retention of color information and distortion on PET/SPECT images. The

third column in Fig. 9 shows the results of FS-Diff at $\times 8$ magnification ($16^2 \rightarrow 128^2$). The fusion results generated by FS-Diff retained excellent edges, contrasts, and structures.

As shown in Table 5, our FS-Diff consistently performed well in terms of the VIF, Q^{AB/F}, SSIM, PSNR, and LPIPS metrics on the MRI-PET and MRI-SPECT MIF datasets slightly lagged behind U2Fusion in terms of the MSE. Notably, although the results of the comparison method were obtained via multimodal HR source image fusion and then down-sampling ($256^2 \rightarrow 128^2$), the subjective and objective assessments were still inferior to the results of FS-Diff ($16^2 \rightarrow 128^2$), which fully proved the effectiveness of FS-Diff on MIF.

5.4. Ablation experiments

5.4.1. The influence of BFM and CLSE

To investigate the effectiveness of BFM and the CLSE mechanism in our proposed method, we conducted an ablation study on the VIF task with MSRS and AVMS datasets. The results are presented in Table 6 and

Table 2

The objective results for infrared and visible image fusion and super resolution ($\times 8$, Red: the best, Blue: the second, Green: the third).

Dataset: MSRS fusion dataset (VI blurred scale: 8)							Dataset: M ² FD fusion dataset (VI and IR blurred scale: 8)						
Methods	VIF $\uparrow$	Q ^{AB/F} $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$	Methods	VIF $\uparrow$	Q ^{AB/F} $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$
CDD+SR3	0.564	0.498	1.204	16.088	0.263	2518.519	CDD+SR3	0.251	0.301	0.892	13.141	0.303	3868.727
Reconet+SR3	0.347	0.323	0.577	15.325	0.318	2024.445	Reconet+SR3	0.226	0.271	0.935	12.024	0.339	3173.556
IGNet+SR3	0.485	0.403	0.978	16.889	0.264	1948.556	IGNet+SR3	0.334	0.321	0.908	12.084	0.281	4362.066
UMF+SR3	0.311	0.264	1.026	15.902	0.247	1582.429	UMF+SR3	0.341	0.302	1.067	13.670	0.246	2427.097
LRRNet+SR3	0.363	0.380	0.979	15.775	0.260	2350.511	LRRNet+SR3	0.175	0.250	0.842	13.028	0.306	3576.946
Tardal+SR3	0.174	0.166	0.231	14.210	0.325	3133.861	Tardal+SR3	0.192	0.233	0.420	10.016	0.448	6472.144
TGFuse+SR3	0.527	0.483	1.207	16.372	0.264	2428.202	TGFuse+SR3	0.232	0.326	0.922	13.006	0.288	3979.542
DDFM+SR3	0.493	0.418	1.226	17.829	0.246	1690.696	DDFM+SR3	0.340	0.307	0.974	14.117	0.268	2841.389
DIF+SR3	0.539	0.483	1.214	16.013	0.250	2593.641	DIF+SR3	0.252	0.320	0.932	12.441	0.306	4406.081
CoCoNet+SR3	0.421	0.290	0.658	9.651	0.376	7186.677	CoCoNet+SR3	0.298	0.245	0.658	11.660	0.381	4497.379
PromptFusion+SR3	0.552	0.495	1.224	16.384	0.260	2436.603	PromptFusion+SR3	0.245	0.289	0.889	13.160	0.305	3788.804
MetaFusion	0.368	0.158	1.119	14.160	0.332	2063.281	MetaFusion	0.191	0.186	0.945	11.169	0.491	3159.876
Our FS-Diff	0.759	0.673	1.277	16.123	0.231	2533.818	Our FS-Diff	0.331	0.364	1.025	12.723	0.287	4278.198

Dataset: AVMS fusion dataset (IR blurred scale: 8)							Dataset: LLVIP fusion dataset (VI blurred scale: 8)						
Methods	VIF $\uparrow$	Q ^{AB/F} $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$	Methods	VIF $\uparrow$	Q ^{AB/F} $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$
CDD+SR3	0.590	0.517	1.306	15.167	0.231	2408.968	CDD+SR3	0.666	0.630	1.190	14.342	0.267	2538.224
Reconet+SR3	0.452	0.431	1.281	14.828	0.315	1816.276	Reconet+SR3	0.369	0.302	0.760	12.449	0.359	2258.865
IGNet+SR3	0.462	0.446	1.096	13.189	0.266	3182.914	IGNet+SR3	0.557	0.562	1.095	15.008	0.272	2090.975
UMF+SR3	0.493	0.455	1.386	15.838	0.214	1386.266	UMF+SR3	0.390	0.368	1.080	14.923	0.276	1460.715
LRRNet+SR3	0.461	0.428	1.195	15.053	0.244	2459.409	LRRNet+SR3	0.348	0.336	0.932	15.576	0.295	1843.225
Tardal+SR3	0.322	0.276	0.648	10.800	0.399	5059.341	Tardal+SR3	0.252	0.294	0.497	13.072	0.413	3117.933
TGFuse+SR3	0.593	0.574	1.312	15.085	0.220	2299.459	TGFuse+SR3	0.555	0.597	1.152	14.834	0.273	2028.494
DDFM+SR3	0.501	0.487	1.344	16.013	0.223	1918.311	DDFM+SR3	0.557	0.541	1.173	14.952	0.268	2423.511
DIF+SR3	0.501	0.527	1.278	14.647	0.241	2546.911	DIF+SR3	0.546	0.562	1.134	13.835	0.293	2538.445
CoCoNet+SR3	0.465	0.350	1.004	13.070	0.297	3222.768	CoCoNet+SR3	0.474	0.419	0.914	11.127	0.320	5318.431
PromptFusion+SR3	0.641	0.540	1.313	15.271	0.226	2462.413	PromptFusion+SR3	0.651	0.622	1.208	14.905	0.262	2272.659
MetaFusion	0.433	0.363	1.127	13.959	0.293	1886.390	MetaFusion	0.242	0.194	0.921	12.476	0.385	2100.118
Our FS-Diff	0.552	0.532	1.344	14.756	0.211	2368.634	Our FS-Diff	0.473	0.481	1.133	13.872	0.259	2511.560

Table 3

The objective results for infrared and visible image fusion and super resolution ($\times 4$, Red: the best, Blue: the second, Green: the third).

Dataset: MSRS fusion dataset (VI blurred scale: 4)							Dataset: M ² FD fusion dataset (VI and IR blurred scale: 4)						
Methods	VIF $\uparrow$	QAB/F $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$	Methods	VIF $\uparrow$	QAB/F $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$
CDD+SR3	0.347	0.278	0.985	15.512	0.293	3344.432	CDD+SR3	0.452	0.374	1.022	11.032	0.267	6187.481
Reconet+SR3	0.323	0.268	0.558	15.262	0.321	2039.084	Reconet+SR3	0.360	0.352	1.121	12.126	0.293	3053.902
IGNet+SR3	0.272	0.210	0.731	14.909	0.374	2234.577	IGNet+SR3	0.215	0.292	0.660	11.288	0.416	4795.159
UMF+SR3	0.294	0.227	1.001	15.882	0.252	1594.493	UMF+SR3	0.405	0.335	1.206	13.473	0.221	2429.874
LRRNet+SR3	0.319	0.303	0.930	15.723	0.266	2381.735	LRRNet+SR3	0.330	0.339	1.048	13.108	0.247	3535.829
Tardal+SR3	0.418	0.264	0.873	12.538	0.282	3733.281	Tardal+SR3	0.453	0.344	1.118	12.951	0.284	3745.289
TGFuse+SR3	0.475	0.396	1.154	16.255	0.270	2477.432	TGFuse+SR3	0.378	0.419	1.131	13.187	0.234	3832.382
DDFM+SR3	0.427	0.328	1.131	17.772	0.272	1696.539	DDFM+SR3	0.414	0.361	1.126	14.295	0.241	2766.588
DIF+SR3	0.500	0.402	1.174	15.927	0.253	2635.370	DIF+SR3	0.399	0.408	1.133	12.596	0.247	4261.666
CoCoNet+SR3	0.391	0.247	0.612	9.622	0.380	7226.883	CoCoNet+SR3	0.445	0.298	0.856	11.886	0.311	4262.548
PromptFusion+SR3	0.495	0.405	1.167	16.282	0.267	2482.416	PromptFusion+SR3	0.444	0.385	1.108	13.174	0.242	3801.288
MetaFusion	0.285	0.219	1.094	15.434	0.299	1916.292	MetaFusion	0.375	0.336	1.093	11.783	0.282	3130.103
Our FS-Diff	0.731	0.669	1.267	16.061	0.233	2528.382	Our FS-Diff	0.583	0.533	1.218	13.440	0.227	3561.357

Dataset: LLVIP fusion dataset (IR blurred scale: 4)							Dataset: AVMS fusion dataset (VI and IR blurred scale: 4)						
Methods	VIF $\uparrow$	QAB/F $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$	Methods	VIF $\uparrow$	QAB/F $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$
CDD+SR3	0.636	0.561	1.137	13.314	0.282	3192.400	CDD+SR3	0.379	0.304	0.969	12.982	0.295	3754.106
Reconet+SR3	0.363	0.295	0.731	12.270	0.361	2359.030	Reconet+SR3	0.356	0.307	1.094	14.372	0.336	1880.610
IGNet+SR3	0.277	0.313	0.777	13.489	0.370	2567.333	IGNet+SR3	0.234	0.250	0.680	11.938	0.406	3747.477
UMF+SR3	0.370	0.334	1.068	14.914	0.279	1463.648	UMF+SR3	0.340	0.279	1.100	14.895	0.253	1563.837
LRRNet+SR3	0.342	0.321	0.911	15.534	0.297	1854.220	LRRNet+SR3	0.307	0.283	0.945	14.736	0.283	2598.644
Tardal+SR3	0.440	0.344	1.053	13.459	0.321	2892.844	Tardal+SR3	0.362	0.294	1.052	14.111	0.322	2849.113
TGFuse+SR3	0.558	0.571	1.144	14.939	0.277	1989.202	TGFuse+SR3	0.339	0.323	1.028	14.754	0.266	2366.751
DDFM+SR3	0.475	0.472	1.114	15.274	0.295	2193.437	DDFM+SR3	0.337	0.304	1.043	15.671	0.267	2002.561
DIF+SR3	0.541	0.554	1.141	14.165	0.292	2312.984	DIF+SR3	0.347	0.320	1.049	14.450	0.277	2590.212
CoCoNet+SR3	0.476	0.400	0.903	11.115	0.322	5340.444	CoCoNet+SR3	0.320	0.245	0.780	12.796	0.328	3415.530
PromptFusion+SR3	0.597	0.569	1.195	15.128	0.266	2148.109	PromptFusion+SR3	0.371	0.303	0.992	14.762	0.275	2683.096
MetaFusion	0.334	0.249	1.025	12.926	0.347	2027.974	MetaFusion	0.340	0.290	1.048	13.467	0.291	2009.857
Our FS-Diff	0.592	0.614	1.179	13.981	0.260	2681.306	Our FS-Diff	0.534	0.503	1.298	15.134	0.229	2405.701

Fig. 10. BFM plays a crucial role in enabling the joint super-resolution and fusion network to perceive and extract global information from different modal images. The CLSE mechanism, on the other hand, can adaptively sense image clarity and extract semantic information from the input images, effectively guiding the fusion process. To evaluate their contributions, we removed the BFM and replaced it with a convolutional layer. Additionally, we directly eliminated the CLSE mechanism to assess its impact on semantic guidance during the fusion process.

The ablation experimental results presented in Table 6 and Fig. 10 demonstrate that the optimal fusion performance is achieved through the synergistic effects of BFM and the CLSE mechanism, thereby validating the rationale behind our FS-Diff framework.

5.4.2. The discussion of semantic selection in CLSE mechanism

We further corroborate the rationality of the CLSE adaptive feature selection rule design. Specifically, when a single-modal image is

blurred, we incorporate the blurred and clear semantics into the fusion process separately. When both modalities are blurred, we embed the blurred single-modal semantics, joint modality semantics, and the maximum of the two-modal semantics into the fusion process, respectively. As shown in Table 7 and Fig. 11, for a single blurred image, choosing the clear semantics and, for two blurred images, selecting the maximum two-modal semantics can effectively enhance fusion performance.

5.4.3. The discussion of zero-shot generation of CLSE mechanism

To evaluate the zero-shot generalization of the adaptive clarity judgment of CLSE, we conducted experiments on the AVMS dataset under compound low-resolution and foggy conditions. In all three experiments, the models did not undergo any prior training and were directly evaluated through inference. As illustrated in Table 8 and Fig. 12, under the unseen weather conditions, the model integrated with the clarity judgment mechanism exhibits superior performance compared to the

Table 4

The objective results for infrared and visible image fusion and super resolution ($\times 2$, Red: the best, Blue: the second, Green: the third).

Dataset: MSRS fusion dataset(VI blurred scale:2)							Dataset: M ² FD fusion dataset(VI and IR blurred scale:2)						
Methods	VIF↑	QAB/F↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓	Methods	VIF↑	QAB/F↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
CDD+SR3	0.322	0.266	1.014	15.283	0.288	3626.851	CDD+SR3	0.391	0.349	0.992	10.860	0.270	6478.874
Reconet+SR3	0.332	0.275	0.687	15.393	0.320	2102.230	Reconet+SR3	0.335	0.339	1.128	12.227	0.298	3112.319
IGNet+SR3	0.258	0.196	0.719	14.744	0.381	2289.869	IGNet+SR3	0.199	0.278	0.645	11.227	0.425	4792.991
UMF+SR3	0.284	0.188	1.057	16.276	0.246	1542.097	UMF+SR3	0.359	0.302	1.178	13.405	0.222	2420.237
LRRNet+SR3	0.286	0.280	0.901	15.419	0.270	2460.521	LRRNet+SR3	0.288	0.318	0.995	12.988	0.252	3609.997
Tardal+SR3	0.391	0.256	0.864	12.444	0.286	3779.646	Tardal+SR3	0.417	0.323	1.093	12.861	0.288	3783.758
TGFuse+SR3	0.401	0.359	1.019	14.803	0.275	2925.301	TGFuse+SR3	0.318	0.379	1.087	12.755	0.235	4138.130
DDFM+SR3	0.410	0.313	1.013	16.790	0.294	1901.532	DDFM+SR3	0.372	0.329	1.090	14.143	0.242	2869.899
DIF+SR3	0.428	0.364	1.058	14.578	0.248	3091.676	DIF+SR3	0.350	0.374	1.102	12.151	0.246	4600.052
CoCoNet+SR3	0.331	0.228	0.555	9.528	0.413	7406.531	CoCoNet+SR3	0.392	0.282	0.797	11.839	0.324	4307.486
PromptFusion+SR3	0.394	0.350	1.070	15.488	0.275	2697.993	PromptFusion+SR3	0.380	0.354	1.052	12.914	0.248	4016.798
MetaFusion	0.287	0.179	1.080	15.232	0.302	1990.558	MetaFusion	0.347	0.328	1.089	11.661	0.277	3225.636
Our FS-Diff	0.834	0.693	1.274	16.085	0.231	2500.553	Our FS-Diff	0.606	0.541	1.226	13.447	0.224	3562.457

Dataset: LLVIP fusion dataset(IR blurred scale:2)							Dataset: AVMS fusion dataset(VI and IR blurred scale:2)						
Methods	VIF↑	QAB/F↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓	Methods	VIF↑	QAB/F↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
CDD+SR3	0.650	0.593	1.156	13.487	0.277	3075.494	CDD+SR3	0.482	0.410	1.134	12.822	0.263	3866.371
Reconet+SR3	0.365	0.297	0.727	12.210	0.364	2386.032	Reconet+SR3	0.433	0.400	1.246	14.684	0.320	1902.222
IGNet+SR3	0.281	0.322	0.784	13.478	0.371	2570.188	IGNet+SR3	0.278	0.308	0.785	12.015	0.390	3687.439
UMF+SR3	0.377	0.340	1.073	14.816	0.279	1446.383	UMF+SR3	0.431	0.382	1.286	15.404	0.220	1475.811
LRRNet+SR3	0.347	0.337	0.905	15.465	0.306	1893.073	LRRNet+SR3	0.378	0.363	1.092	14.920	0.252	2501.392
Tardal+SR3	0.447	0.354	1.062	13.489	0.320	2851.051	Tardal+SR3	0.435	0.378	1.197	14.190	0.299	2818.125
TGFuse+SR3	0.576	0.597	1.159	14.957	0.272	1947.154	TGFuse+SR3	0.440	0.451	1.207	14.676	0.232	2482.844
DDFM+SR3	0.484	0.488	1.125	15.345	0.295	2152.899	DDFM+SR3	0.423	0.410	1.240	15.933	0.232	1933.640
DIF+SR3	0.534	0.560	1.115	13.586	0.295	2662.501	DIF+SR3	0.450	0.447	1.217	14.278	0.249	2751.751
CoCoNet+SR3	0.484	0.413	0.912	11.123	0.320	5329.684	CoCoNet+SR3	0.388	0.306	0.917	12.974	0.310	3288.426
PromptFusion+SR3	0.617	0.595	1.209	15.201	0.263	2125.803	PromptFusion+SR3	0.481	0.417	1.180	14.921	0.242	2611.946
MetaFusion	0.351	0.295	1.015	13.601	0.341	1782.251	MetaFusion	0.424	0.383	1.201	13.862	0.269	1975.047
Our FS-Diff	0.607	0.621	1.182	13.953	0.258	2704.236	Our FS-Diff	0.550	0.510	1.307	15.163	0.222	2390.913

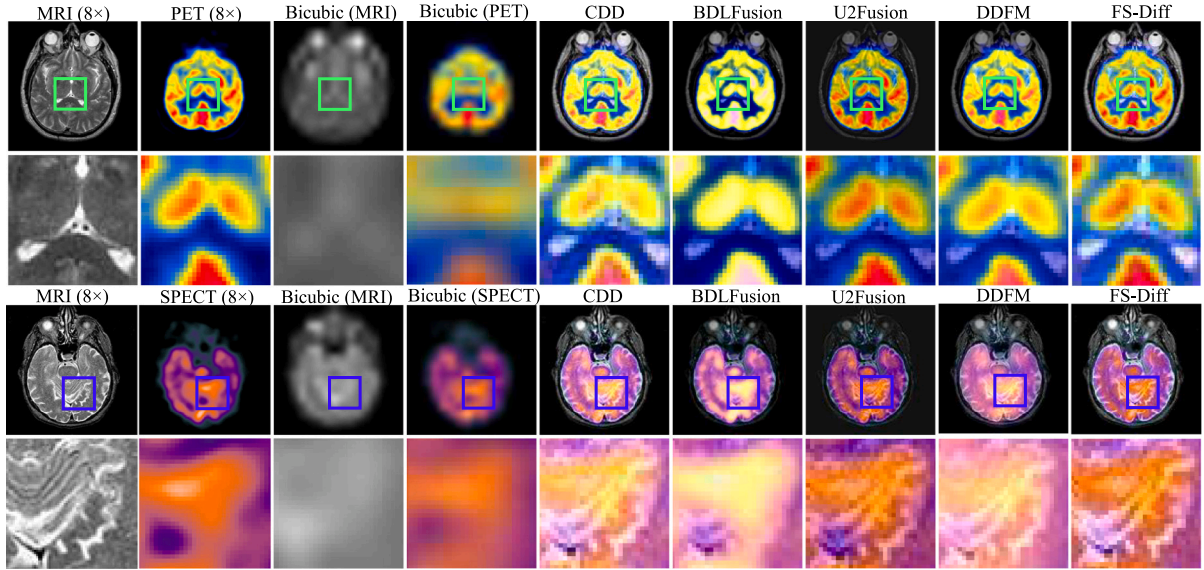


Fig. 9. Fusion and super-resolution (scale:8) results for FS-Diff as well as the fusion results for the compared methods on the Harvard medical image dataset.

Table 5

The objective results for medical image fusion and super resolution (Red: the best, Blue: the second best).

Dataset: MRI-PET of Harvard medical image dataset							
Methods	Scale	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
CDD	1	0.714	0.690	1.335	12.889	0.303	3505.219
BDLFusion	1	0.479	0.528	1.154	13.782	0.308	2525.134
U2Fusion	1	0.481	0.415	0.620	14.550	0.293	1874.329
DDFM	1	0.659	0.594	1.285	14.614	0.292	2057.891
Our FS-Diff	8	0.760	0.719	1.377	13.367	0.303	3194.768

Dataset: MRI-SPECT of Harvard medical image dataset							
Methods	Scale	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
CDD	1	0.709	0.642	1.468	17.115	0.228	1290.771
BDLFusion	1	0.474	0.388	1.361	16.259	0.231	1488.426
U2Fusion	1	0.412	0.358	0.613	17.392	0.226	729.343
DDFM	1	0.552	0.560	1.418	16.966	0.237	1289.635
Our FS-Diff	8	0.686	0.639	1.476	18.057	0.224	1022.069

Table 6

Ablation experiment on the CLSE and BFM. (Red: the best).

Method									Method								
CLSE		BFM	VIF↑	QAB/F↑	MSRS (both blurred) scale:8				CLSE		BFM	VIF↑	QAB/F↑	AVMS (both blurred) scale:8			
				SSIM↑	PSNR↑	LPIS↓	MSE↓						SSIM↑	PSNR↑	LPIS↓	MSE↓	
1	✓	✗	0.472	0.479	1.061	13.489	0.343	2662.481	1	✓	✗	0.283	0.248	0.862	13.817	0.423	2974.492
2	✗	✓	0.479	0.471	1.055	13.592	0.299	2757.148	2	✗	✓	0.257	0.283	0.846	13.574	0.367	2967.923
3	✗	✗	0.453	0.461	0.848	12.489	0.328	2904.463	3	✗	✗	0.267	0.198	0.581	12.489	0.593	3049.596
Our FS-Diff	✓	✓	0.737	0.669	1.285	16.170	0.229	2496.989	Our FS-Diff	✓	✓	0.533	0.516	1.332	14.716	0.211	2370.454
Method									Method								
CLSE		BFM	VIF↑	QAB/F↑	MSRS (VI blurred) scale:8				CLSE		BFM	VIF↑	QAB/F↑	AVMS (VI blurred) scale:8			
				SSIM↑	PSNR↑	LPIS↓	MSE↓						SSIM↑	PSNR↑	LPIS↓	MSE↓	
1	✓	✗	0.456	0.567	1.199	15.567	0.339	2949.577	1	✓	✗	0.513	0.445	1.213	9.592	0.322	4365.434
2	✗	✓	0.623	0.534	1.159	14.533	0.317	2694.419	2	✗	✓	0.482	0.442	1.151	11.192	0.426	5423.461
3	✗	✗	0.328	0.598	1.012	12.962	0.511	3359.489	3	✗	✗	0.449	0.297	1.048	10.495	0.311	5119.189
Our FS-Diff	✓	✓	0.771	0.683	1.288	16.172	0.229	2500.538	Our FS-Diff	✓	✓	0.546	0.523	1.336	14.744	0.211	2374.510
Method									Method								
CLSE		BFM	VIF↑	QAB/F↑	MSRS (IR blurred) scale:8				CLSE		BFM	VIF↑	QAB/F↑	AVMS (IR blurred) scale:8			
				SSIM↑	PSNR↑	LPIS↓	MSE↓						SSIM↑	PSNR↑	LPIS↓	MSE↓	
1	✓	✗	0.412	0.583	1.009	15.277	0.399	2919.504	1	✓	✗	0.466	0.479	1.304	13.489	0.298	2548.192
2	✗	✓	0.647	0.611	1.068	15.192	0.308	2715.189	2	✗	✓	0.492	0.433	1.239	13.994	0.331	2615.919
3	✗	✗	0.348	0.556	0.963	13.489	0.392	3000.489	3	✗	✗	0.348	0.358	1.095	13.496	0.469	2948.953
Our FS-Diff	✓	✓	0.761	0.678	1.287	16.176	0.229	2500.053	Our FS-Diff	✓	✓	0.543	0.522	1.335	14.736	0.211	2373.633

Table 7

The impact of content embeddings of different image modalities on fusion performance. (Red: the best).

Models	Dataset: AVMS fusion dataset (scale:8)					
	VIF↑	QAB/F↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
VI clear + IR semantics	0.384	0.301	1.150	14.127	2377.841	0.419
VI clear + VI semantics	0.543	0.522	1.335	14.736	2373.633	0.211
IR clear + VI semantics	0.385	0.303	1.153	14.145	2378.165	0.422
IR clear + IR semantics	0.546	0.523	1.336	14.744	2374.510	0.211
Blurred IR & VI + IR semantics	0.423	0.365	1.217	14.331	2371.607	0.354
Blurred IR & VI + VI semantics	0.414	0.350	1.204	14.270	2371.986	0.362
Blurred IR & VI + joint semantics	0.384	0.301	1.151	14.143	2377.909	0.421
Blurred IR & VI + max semantics	0.533	0.516	1.332	14.716	2370.454	0.211

Table 8

The impact of CLSE's clarity judgment on the unseen foggy and low resolution AVMS Dataset. (Red: the best).

Structure of CLSE	Dataset: AVMS fusion dataset (foggy & LR (scale:8))					
	VIF↑	QAB/F↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
w/o clarity judgment	0.379	0.293	1.105	9.920	5908.992	0.375
w VGG19	0.348	0.257	1.081	10.684	4902.755	0.434
w clarity judgment	0.508	0.497	1.319	13.486	2943.330	0.218

model lacking this mechanism and the model substituted with the pre-trained VGG19 network. This cross-condition performance improvement, achieved without specific training, underscores the necessity of the clarity judgment of CLSE.

5.5. Performance on advanced visual task

To validate the flexibility of FS-Diff in advanced vision tasks, we conducted experiments on two tasks, namely, image segmentation and detection, using public datasets containing information about larger targets and our AVMS dataset, which was enriched with small targets.

5.5.1. Comparisons on object detection

Target detection experiments were conducted on the LLVIP and AVMS datasets, via pretrained YOLOv8 [118] as the backbone. Fig. 13 shows the results of the qualitative analysis. All the targets in the unimodal images were missed, and mis-detected regions were noted in the comparison methods. By contrast, FS-Diff shows superior detection for the targets of “people” and “cars”, obtaining a high confidence score, owing to its rich semantic features. For quantitative analysis, from Table 9, FS-Diff shows the best detection performance for mAP@0.5, mAP@0.75, and mAP@0.5:0.95, followed closely by LRRNet.

5.5.2. Comparisons on segmentation task

Image segmentation experiments were conducted on the MSRS and our AVMS datasets, using BiSeNet [119] retrained on the MSRS as the backbone of the segmentation task. The qualitative analysis shown in Fig. 14 shows that the segmentation results of unimodal LR images are generally worse than the fusion results. In addition, the separate processes of fusion and super-resolution lead to the loss of semantic information and the accumulation of errors, causing the comparison

methods to fail in preserving segmentation regions. In contrast, FS-Diff produces more accurate segmentation results. For example, in the MSRS dataset, other methods struggle to segment “people” accurately, whereas FS-Diff effectively highlights the hot target information. Moreover, CDD, Reconet, IGNet, Tardal, DIF and CoCoNet have difficulty in accurately segmenting “buses” in AVMS. Although FS-Diff fails to completely segment all vehicles on the road, it segments the most semantic information, compared with other methods.

The quantitative analyses are shown in Table 10. The mIoU values of FS-Diff on the MSRS and AVMS datasets were 3.596% and 2.151% higher than those of the second-ranked DDFM and IGNet, respectively. These results show that FS-Diff exhibits superior performance in image segmentation tasks.

5.6. Extended experiments

Existing multi-focus image fusion methods [23,34,120–123] are limited to fixed resolutions. Although strategies such as performing super-resolution prior to fusion can be employed, the former may introduce artifacts that, when amplified in subsequent steps, significantly degrade the final result. Simultaneous super-resolution and fusion of multifocus images enhance the information content in the fused result, preserving more focused features from the source images. To further validate the generalizability of FS-Diff, we compared it with Fusion-Diff [120], RRSTD [34], MUFusion [121], and U2Fusion [23] on the public datasets Lytro [122] and MFI [123]. The stepwise fusion and super-resolution methods were used for comparison. For training, HR fusion images were obtained via U2Fusion [23] and training image pairs were obtained by bicubically downsampling the source images. To be consistent with the above experiment, we centered and cropped

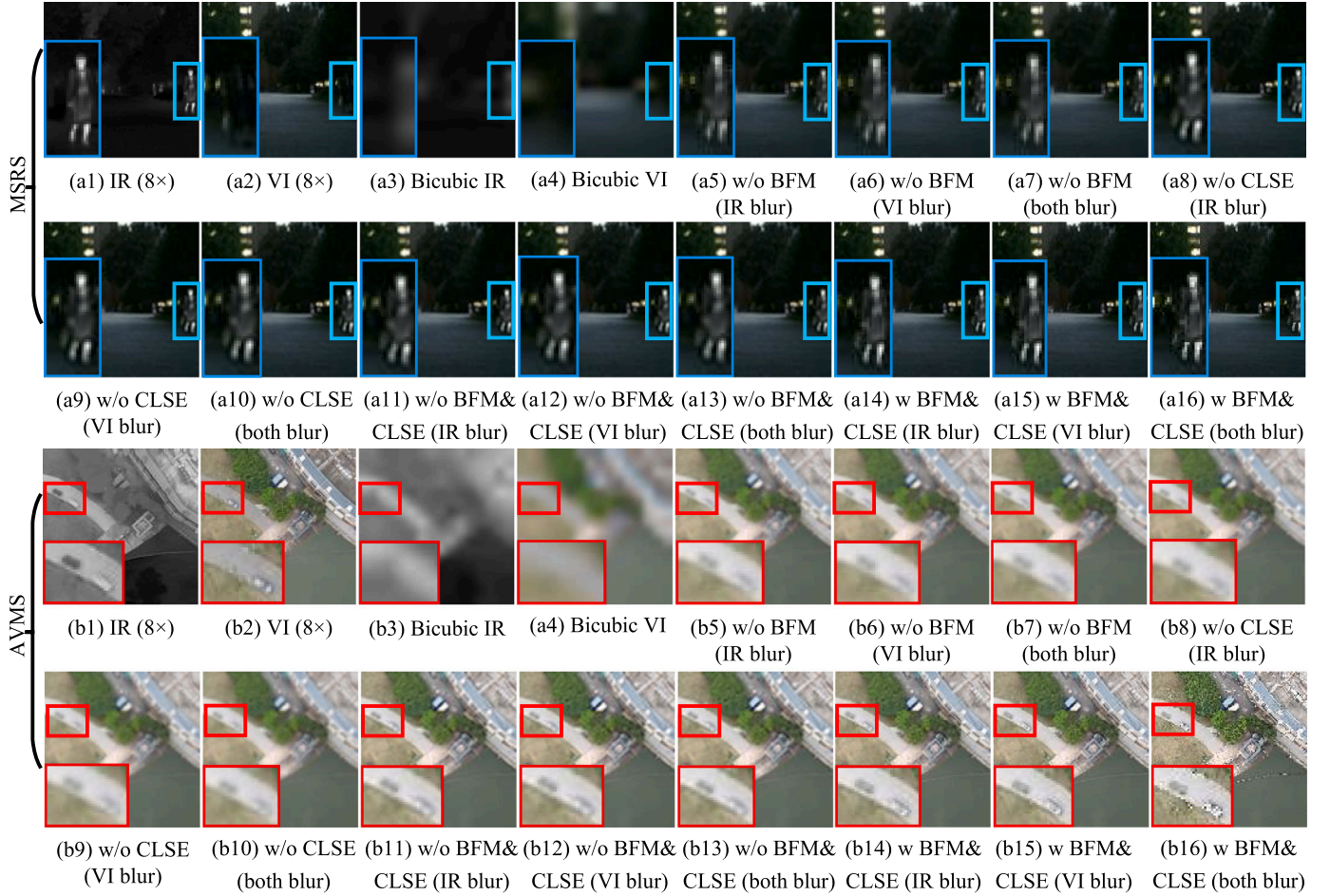


Fig. 10. Visual comparison of ablation studies on CA-CLIP and CLSE Mechanism.

Table 9

Quantitative results of objective detection on LLVIP and AVMS datasets. (Red: the best, Blue: the second best).

Methods	Dataset: LLVIP			Dataset: AVMS		
	mAP@0.50	mAP@0.75	mAP@0.50:0.95	mAP@0.50	mAP@0.75	mAP@0.50:0.95
CDD+SR3	0.832	0.633	0.581	0.750	0.680	0.493
Reconet+SR3	0.886	0.671	0.594	0.773	0.622	0.510
IGNet+SR3	0.899	0.684	0.578	0.783	0.508	0.488
UMF+SR3	0.895	0.658	0.586	0.782	0.614	0.580
LRRNet+SR3	0.927	0.716	0.604	0.733	0.685	0.573
Tardal+SR3	0.893	0.692	0.609	0.773	0.627	0.545
TGFuse+SR3	0.913	0.702	0.592	0.756	0.617	0.447
DDFM+SR3	0.851	0.623	0.515	0.740	0.522	0.462
DIF+SR3	0.882	0.653	0.532	0.723	0.647	0.440
CoCoNet+SR3	0.871	0.638	0.548	0.762	0.638	0.491
PromptFusion+SR3	0.868	0.693	0.551	0.772	0.651	0.487
Our FS-Diff	0.933	0.720	0.624	0.795	0.728	0.598

the two datasets, degraded them to 16×16 , and termed them Lytro-val and MFI-val, respectively.

The quantitative evaluation under the condition where both image modalities are blurred in Table 11 shows that FS-Diff achieved the best values for the VIF, $Q^{AB/F}$, SSIM and LPIPS metrics, with RRSTD closely following. Compared with the SR3-only (far/near) method, which applies super-resolution to the two source images, FS-Diff has significant advantages in preserving visual information fidelity, structural similarity, fusion quality, and visual consistency. As shown in Fig. 15, despite the super-resolution processing of the low-resolution image (Figs. 15 (a3)-(a4), (b3)-(b4)), the out-of-focus information remains

arduous to recover (Figs. 15 (a5)-(a6), (b5)-(b6)). For a more intuitive visual evaluation, Figs. 16 (c2)-(g2) and (j8)-(n2) show the differences obtained using the different methods. Less information in the difference map is obtained by subtracting the focus area in one source image from the fusion image, implying that more focus information is extracted by the fusion method. The enlarged area in the lower right corner of Fig. 16, FusionDiff, MUFusion, NSST, and U2Fusion have different levels of residuals around the edge of the hat of the “child” and around the ears of the “lady with the umbrella”. In addition, FusionDiff’s stepwise fusion and super-resolution for the “lady with the umbrella” is unsatisfactory, with a severe loss of semantic information. By contrast,

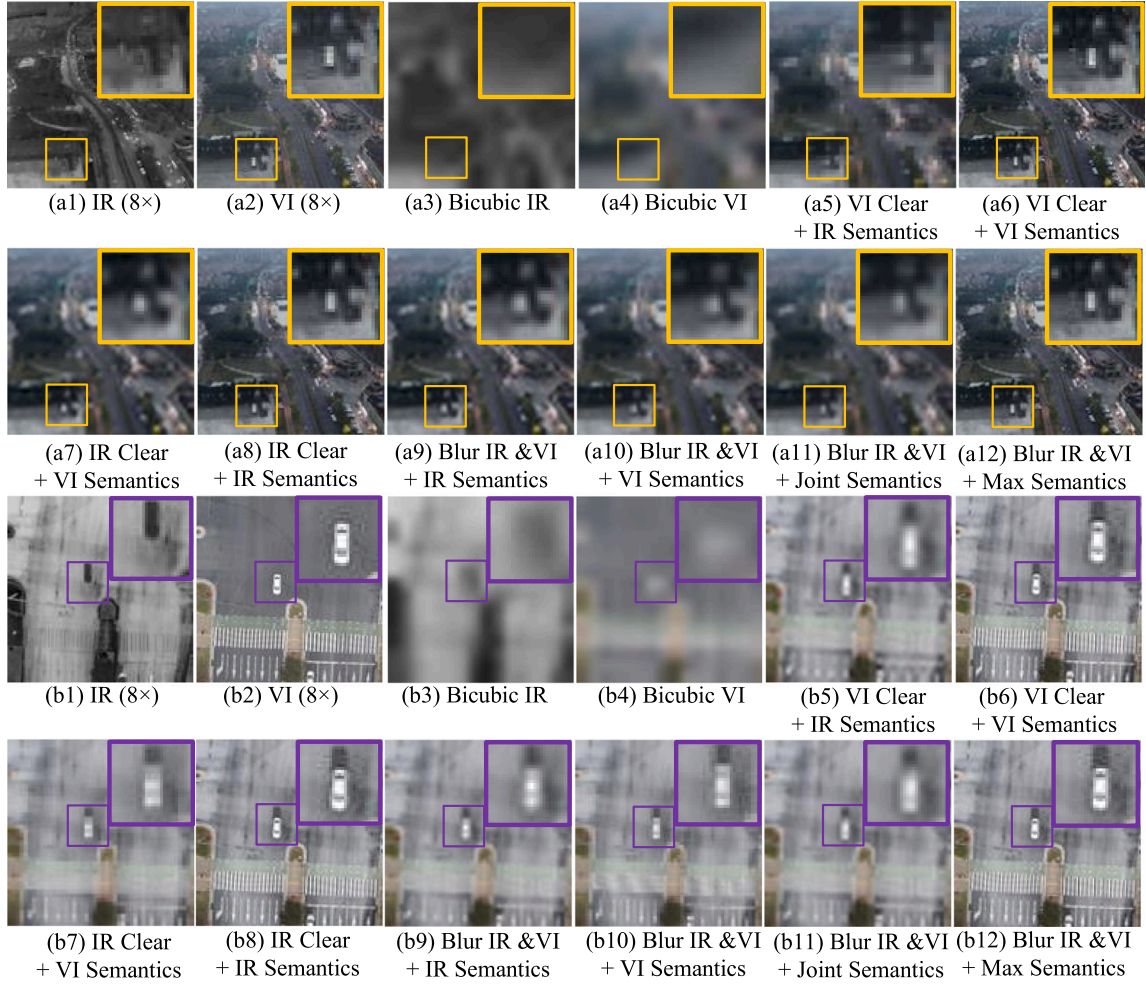


Fig. 11. The impact of content embeddings from images in different modalities for the fusion results.

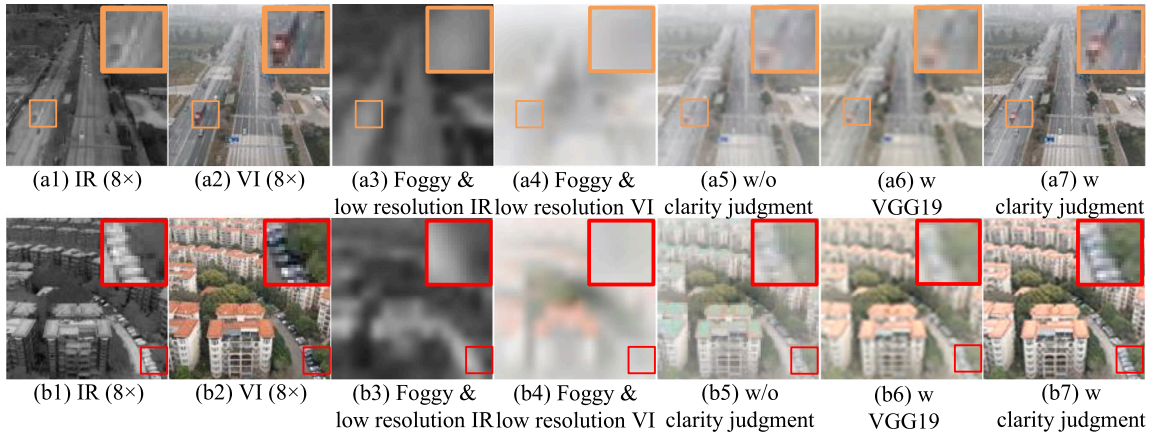


Fig. 12. Visualization of the effect of CLSE's clarity judgment on the unseen foggy and low resolution (scale:8) AVMS dataset.

Figs. 16 (g2) and (n2) show the most continuous and cleanest hat and ear edges, respectively, and preserve the continuous boundary information.

6. Computational complexity analysis

This section compares the computational efficiency and memory consumption between our proposed method and the compared methods. The FLOPs and parameters are quantified at the input size of

128×128 , whereas the running time is averaged over ten 128×128 images from the AVMS dataset. From Table 12, the proposed FS-Diff method has parameter of 58.47M, which is slightly greater than methods like Reconet, UMF and Tardal, but still moderate compared to LRRNet and DDFM. In terms of computational complexity, FS-Diff significantly higher than CDD and Reconet, primarily because of the simultaneous execution of image super-resolution and fusion tasks. Despite the increased computational cost and inference time, FS-Diff delivers finer image details and superior fusion quality. Furthermore,

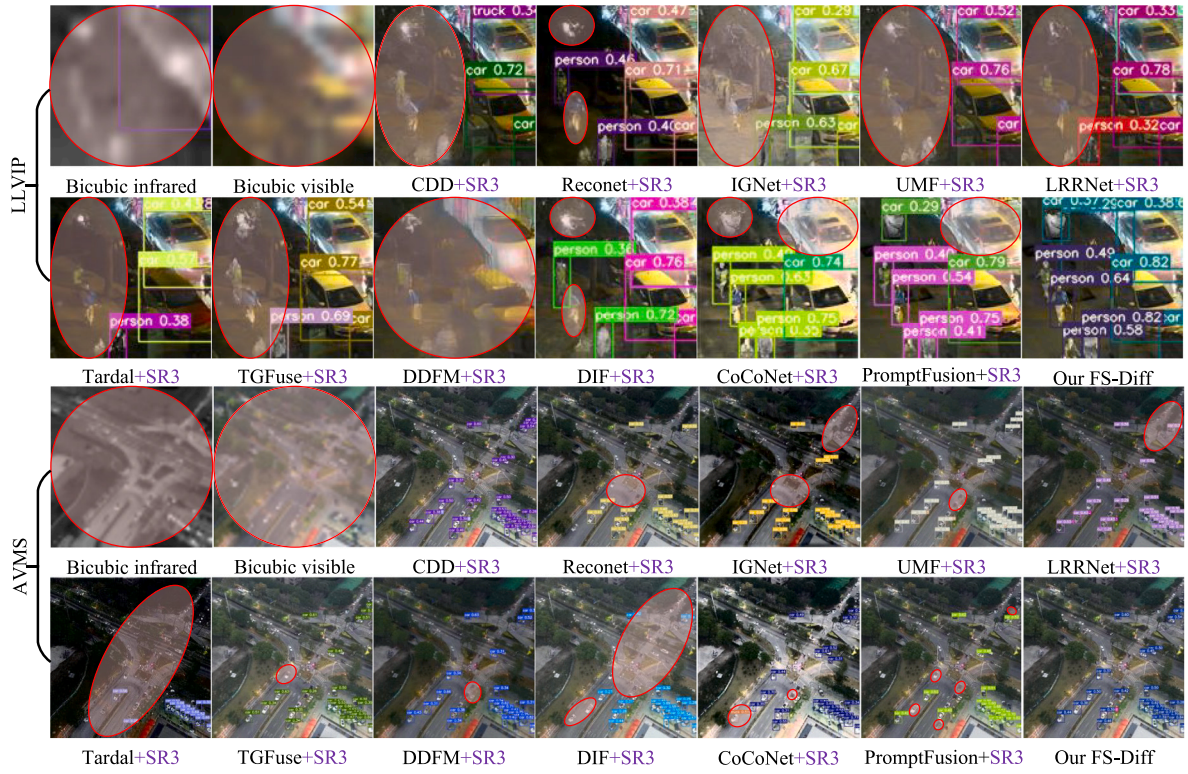


Fig. 13. Qualitative comparisons of different fusion and super-resolution (scale: 8) detection results on the LLVIP and AVMS datasets. The area circled by the red ellipse indicates the missed detection of the comparison methods compared with our FS-Diff.

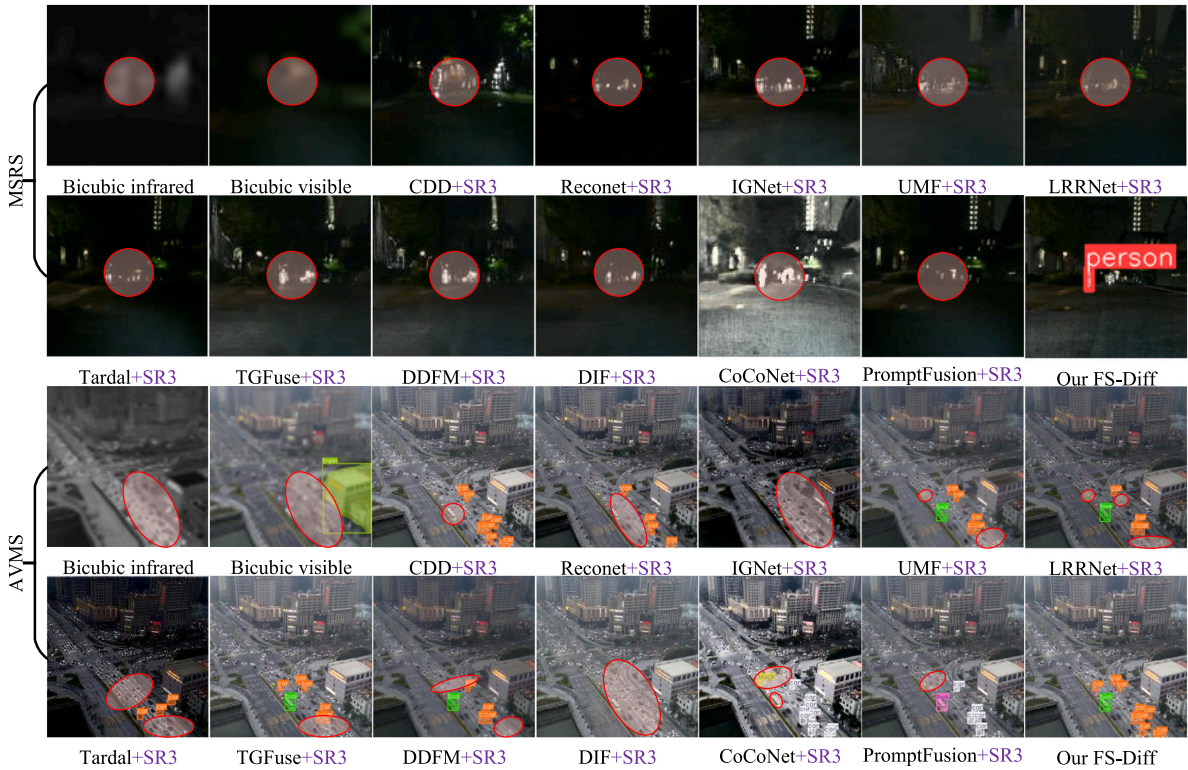


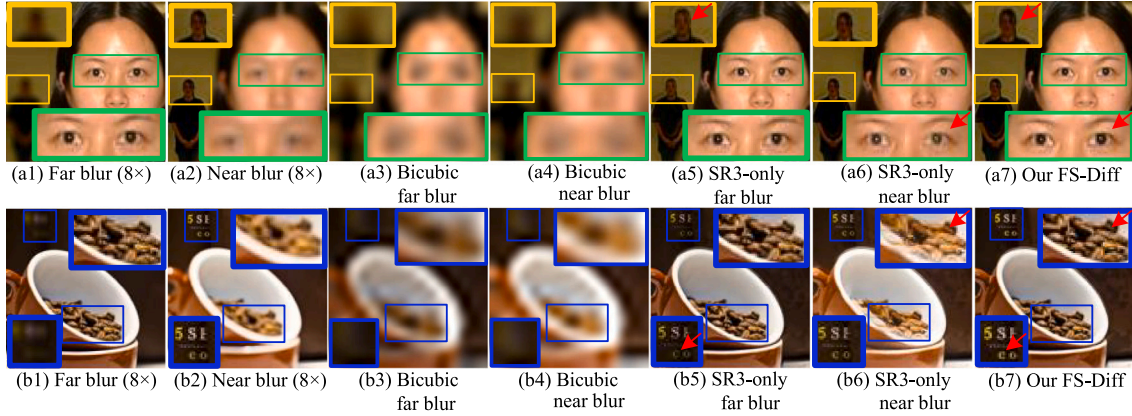
Fig. 14. Qualitative comparisons of different fusion and super-resolution (scale: 8) segmentation results on the MSRS and AVMS datasets. The red regions represent the missing segmentations of the comparison methods compared with our FS-Diff.

Table 10

Quantitative results of segmentation on MSRS and AVMS datasets. (Red: the best, Blue: the second best).

Methods	Dataset: MSRS									
	Background	Car	Person	Bike	Curve	Car stop	Guardrail	Color tone	Bump	mIoU
CDD+SR3	93.057	79.195	69.879	62.939	50.165	64.259	79.475	67.937	71.707	0.710
Reconet+SR3	92.856	67.922	70.125	69.354	57.234	54.216	78.244	68.129	57.242	0.684
IGNet+SR3	92.852	78.684	77.337	68.555	63.552	62.729	83.242	66.843	43.686	0.708
UMF+SR3	92.73	74.861	71.165	56.371	51.462	51.451	60.465	62.771	63.729	0.650
LRRNet+SR3	92.953	79.078	71.545	71.291	63.462	64.574	77.251	67.375	53.353	0.712
Tardal+SR3	92.589	62.474	70.087	65.239	61.368	61.577	78.465	66.033	72.467	0.700
TGFuse+SR3	93.093	80.447	69.735	62.331	54.327	55.088	72.461	66.305	72.614	0.696
DDFM+SR3	93.551	78.251	75.259	69.267	58.247	62.457	78.366	64.251	71.250	0.723
DIF+SR3	92.493	72.117	60.644	64.827	48.428	61.537	73.496	59.242	62.591	0.662
CoCoNet+SR3	92.489	79.415	79.546	68.367	62.189	51.159	71.156	61.524	67.128	0.699
PromptFusion+SR3	93.195	72.156	72.235	76.951	57.462	57.416	78.954	66.554	71.089	0.717
Our FS-Diff	93.217	81.561	77.341	69.846	63.627	64.968	82.523	68.951	72.961	0.749

Methods	Dataset: AVMS									
	Background	Car	Person	Bike	Truck	Electric bicycle	Tent	Grab	Bus	mIoU
CDD+SR3	0.980	0.734	0.803	0.769	0.585	0.445	0.609	0.156	0.718	0.644
Reconet+SR3	0.946	0.719	0.773	0.711	0.577	0.654	0.557	0.167	0.682	0.643
IGNet+SR3	0.979	0.689	0.889	0.768	0.489	0.585	0.660	0.106	0.693	0.651
UMF+SR3	0.907	0.865	0.667	0.714	0.596	0.522	0.519	0.111	0.698	0.622
LRRNet+SR3	0.972	0.844	0.675	0.742	0.571	0.437	0.693	0.138	0.627	0.633
Tardal+SR3	0.947	0.864	0.778	0.530	0.572	0.642	0.404	0.104	0.506	0.594
TGFuse+SR3	0.905	0.774	0.772	0.776	0.596	0.571	0.508	0.178	0.590	0.630
DDFM+SR3	0.967	0.854	0.681	0.610	0.529	0.660	0.536	0.116	0.650	0.623
DIF+SR3	0.933	0.810	0.821	0.631	0.441	0.449	0.617	0.123	0.787	0.623
CoCoNet+SR3	0.956	0.824	0.862	0.751	0.492	0.427	0.546	0.246	0.613	0.635
PromptFusion+SR3	0.948	0.831	0.857	0.734	0.490	0.413	0.551	0.257	0.603	0.632
Our FS-Diff	0.976	0.869	0.875	0.824	0.511	0.438	0.657	0.223	0.613	0.665

**Fig. 15.** Visual Comparison of SR3-only super-resolution and the proposed method for simultaneous multi-source image fusion and super-resolution (scale:8).**Table 11**

The objective results for multi-focus image fusion and super resolution (Red: the best, Blue: the second best).

Methods	Dataset: Lytro (scale:8)					
	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
FusionDiff+SR3	0.974	0.648	1.642	22.387	0.114	742.285
RRSTD+SR3	1.136	0.744	1.787	27.449	0.076	205.217
MUFusion+SR3	1.112	0.735	1.784	25.360	0.077	240.048
U2Fusion+SR3	1.030	0.723	1.755	23.683	0.080	338.688
SR3-only(far)	1.091	0.723	1.783	27.635	0.078	192.624
SR3-only(near)	1.112	0.734	1.787	27.635	0.076	198.685
Our FS-Diff	1.269	0.779	1.797	27.384	0.074	221.278

Methods	Dataset: MFI (scale:8)					
	VIF↑	Q ^{AB/F} ↑	SSIM↑	PSNR↑	LPIPS↓	MSE↓
FusionDiff+SR3	0.965	0.635	1.681	21.979	0.102	713.968
RRSTD+SR3	1.365	0.832	1.937	33.401	0.029	30.820
MUFusion+SR3	1.344	0.827	1.928	27.444	0.031	122.392
U2Fusion+SR3	1.267	0.819	1.913	26.634	0.033	172.695
SR3-only(far)	1.362	0.831	1.937	33.436	0.029	30.578
SR3-only(near)	1.361	0.831	1.937	33.420	0.029	30.612
Our FS-Diff	1.442	0.855	1.941	32.266	0.028	60.797

when adding the SR3 parameters, the FLOPs and inference time (where $n = 1$ when a single image is blurred and $n = 2$ when both images are

blurred) are used for separately performing super-resolution and fusion tasks for the compared methods, the overall computational burden far exceeds that of FS-Diff.

7. Conclusion

To address the resolution damage to image targets and backgrounds caused by long-distance shooting and the limitations of single imaging devices in fully capturing scene information because of sensor constraints, we proposed FS-Diff, a unified image super-resolution and fusion framework. By presenting the CLSE mechanism with CA-CLIP, FS-Diff achieves adaptive clarity awareness for both single- and dual-source images with reduced resolution. It also provides semantic guidance during the fusion process. Additionally, we introduced the BFM to enhance the global feature extraction of multimodal images. Taking LR multimodal images and semantics guidance as conditions, we generated HR-fused results through an iterative refinement denoising process in an end-to-end manner, effectively preserving the high-fidelity detail and cross-modal information required for high-magnification fused images. We conducted extensive experiments on three image fusion tasks: VIRF, MIF, and multifocus image fusion. The results of the quantitative and qualitative evaluations fully demonstrated the strong robustness of

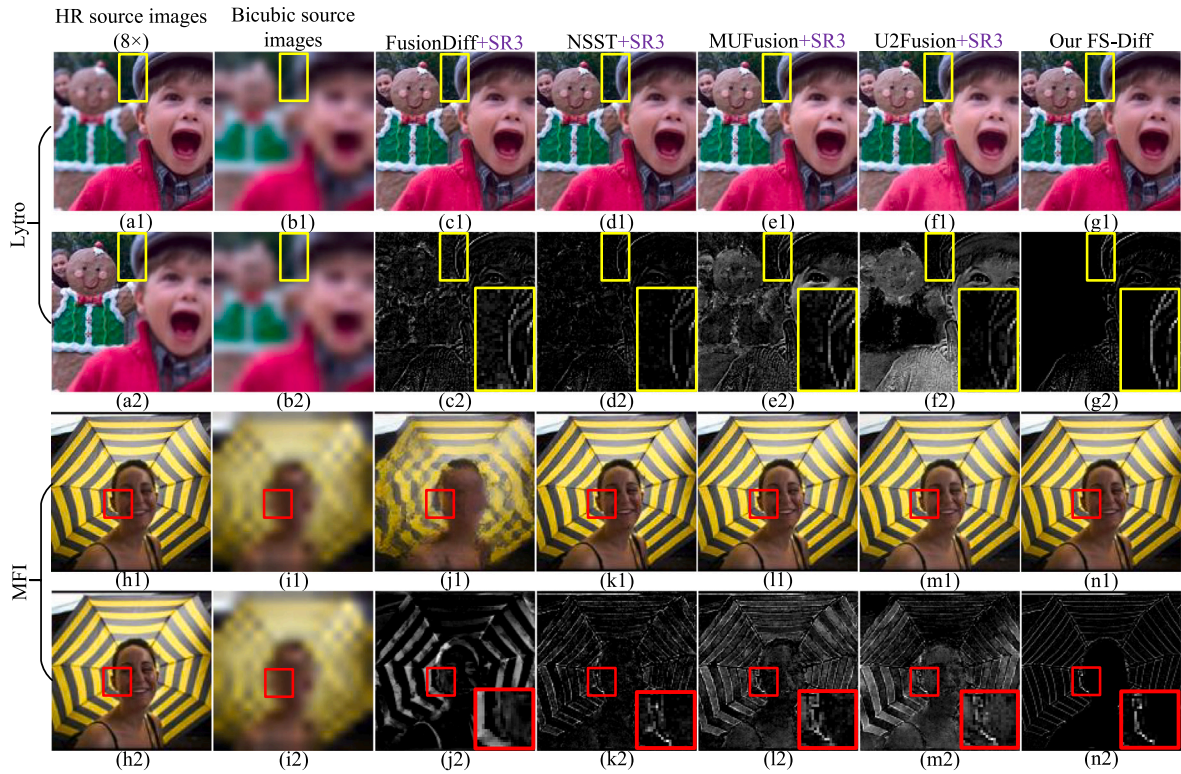


Fig. 16. Fusion and super-resolution (scale: 8) results for different methods on the Lytro and MFI multi-focus datasets. (a1)–(a2) and (h1)–(h2) are partially focused source images; (b1)–(b2) and (i1)–(i2) are input images obtained via bicubic downsampling; (c1)–(g1) and (j1)–(n1) are fused results; (c2)–(g2) are the differences produced by (c1)–(g1) with (a2); and (j2)–(n2) are the differences produced by (j1)–(n1) with (h2).

Table 12

Comparison of computational complexity and inference time.

Methods	CDD+SR3	Reconet+SR3	IGNet+SR3	UMF+SR3	LRRNet+SR3	Tardal+SR3	TGFuse+SR3
Parameters(M)	1.19+97.76	0.01+97.76	7.87+97.76	0.63+97.76	755.57+97.76	0.30+97.76	137.34+97.76
FLOPs(G)	29.21+91681.6	1.52+91681.6	13.49+91681.6	10.30+91681.6	4.90×10^{-5} +91681.6	38.89+91681.6	3.99+91681.6
Time(s)	0.017+ $n \times 35$	0.002+ $n \times 35$	0.005+ $n \times 35$	0.002+ $n \times 35$	0.009+ $n \times 35$	0.087+ $n \times 35$	0.013+ $n \times 35$
Methods	DDFM+SR3	DIF+SR3	Diff-IF+SR3	CoCoNet+SR3	PromptFusion+SR3	MetaFusion	FS-Diff
Parameters(M)	552.81+97.76	32.20+97.76	23.71+97.76	9.13+97.76	7.44+97.76	0.75	58.47
FLOPs(G)	1114.71+91681.6	14.97+91681.6	43.53+91681.6	10.39+91681.6	44.56+91681.6	97.18	64,391.60
Time(s)	34.50+ $n \times 35$	0.072+ $n \times 35$	0.290+ $n \times 35$	0.015+ $n \times 35$	0.189+ $n \times 35$	0.077	74.0

FS-Diff in joint fusion and super-resolution tasks with multiple high-magnification ratios as well as its application value in advanced visual tasks. In addition, to broaden the application boundaries of image fusion methods from an aerial photography perspective, we constructed an AVMS dataset. It includes a variety of lighting intensities, scenes, and 8991 annotations for objects. This dataset can be used for MMIF, image super-resolution, object detection, and semantic segmentation tasks, providing a new benchmark for research in computer vision.

Although the FS-Diff framework demonstrated significant effects on joint image super-resolution and fusion, it has limitations such as high model complexity and long running times. In the future, we plan to improve the U-Net structure of our model to reduce model complexity and further enhance the ability of FS-Diff to adapt quickly to complex and variable real-world scenarios. Simultaneously, we are integrating further research on FS-Diff into autonomous driving systems, UAV precision agricultural irrigation technology, and UAV military combat strategies. This will expand its applications and practices in various fields, including intelligent transportation, agricultural modernization, and defense science and technology.

CRediT authorship contribution statement

Yuchan Jie: Writing – original draft, Validation, Software, Methodology, Data curation. **Yushen Xu:** Validation, Software, Data curation,

Conceptualization. **Xiaosong Li:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Fuqiang Zhou:** Writing – review & editing. **Jianming Lv:** Writing – review & editing. **Huafeng Li:** Writing – review & editing, Methodology.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors have approved the manuscript that is enclosed.

Acknowledgments

This research was supported by the National Natural Science Foundation of China (No. 62201149), the Natural Science Foundation of Guangdong Province (No. 2024A1515011880), the Basic and Applied Basic Research of Guangdong Province (Nos. 2023A1515140077, 2024A1515012287), the Science and Technology Key Program of Guangzhou (No. 2023B03J1388), the National Key R&D Program of China (No. 2023YFA1011601), and the Yunnan Fundamental Research Projects (Nos. 202301AV070004, 202501AS070123).

Data availability

Data will be made available on request.

References

- [1] H. Xu, J. Yuan, J. Ma, Murf: Mutually reinforcing multi-modal image registration and fusion, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [2] Z. Zhao, H. Bai, Y. Zhu, J. Zhang, S. Xu, Y. Zhang, K. Zhang, D. Meng, R. Timofte, L. Van Gool, DDFM: denoising diffusion model for multi-modality image fusion, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 8082–8093.
- [3] W. Zhao, S. Xie, F. Zhao, Y. He, H. Lu, Metafusion: Infrared and visible image fusion via meta-feature embedding from object detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13955–13965.
- [4] C. He, K. Li, G. Xu, Y. Zhang, R. Hu, Z. Guo, X. Li, Degradation-resistant unfolding network for heterogeneous image fusion, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 12611–12621.
- [5] H. Li, X.-J. Wu, CrossFuse: A novel cross attention mechanism based infrared and visible image fusion approach, *Inf. Fusion* 103 (2024) 102147.
- [6] L. Tang, Y. Deng, Y. Ma, J. Huang, J. Ma, SuperFusion: A versatile image registration and fusion network with semantic awareness, *IEEE/CAA J. Autom. Sin.* 9 (12) (2022) 2121–2137.
- [7] S.M.Z.S.Z. Ariffin, N. Jamil, P.N.M.A. Rahman, Can thermal and visible image fusion improves ear recognition? in: *2017 8th International Conference on Information Technology, ICIT, IEEE*, 2017, pp. 780–784.
- [8] R. Liu, Z. Liu, J. Liu, X. Fan, Searching a hierarchically aggregated fusion architecture for fast multi-modality image fusion, in: *Proceedings of the 29th ACM International Conference on Multimedia*, 2021, pp. 1600–1608.
- [9] C. He, K. Li, Y. Zhang, L. Tang, Y. Zhang, Z. Guo, X. Li, Camouflaged object detection with feature decomposition and edge reconstruction, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 22046–22055.
- [10] H. Li, X.-J. Wu, J. Kittler, MDLatLRR: A novel decomposition method for infrared and visible image fusion, *IEEE Trans. Image Process.* 29 (2020) 4733–4746.
- [11] Y. Jie, X. Li, H. Tan, F. Zhou, G. Wang, Multi-modal medical image fusion via multi-dictionary and truncated huber filtering, *Biomed. Signal Process. Control.* 88 (2024) 105671.
- [12] Y. Jie, X. Li, F. Zhou, H. Tan, et al., Medical image fusion based on extended difference-of-Gaussians and edge-preserving, *Expert Syst. Appl.* 227 (2023) 120301.
- [13] X. Zhang, Y. Demiris, Visible and infrared image fusion using deep learning, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [14] S. Karim, G. Tong, J. Li, A. Qadir, U. Farooq, Y. Yu, Current advances and future perspectives of image fusion: A comprehensive review, *Inf. Fusion* 90 (2023) 185–217.
- [15] H. Li, Y. Cen, Y. Liu, X. Chen, Z. Yu, Different input resolutions and arbitrary output resolution: A meta learning-based deep framework for infrared and visible image fusion, *IEEE Trans. Image Process.* 30 (2021) 4070–4083.
- [16] W. Xiao, Y. Zhang, H. Wang, F. Li, H. Jin, Heterogeneous knowledge distillation for simultaneous infrared-visible image fusion and super-resolution, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–15.
- [17] H. Yin, S. Li, L. Fang, Simultaneous image fusion and super-resolution using sparse representation, *Inf. Fusion* 14 (3) (2013) 229–240.
- [18] H. Li, M. Yuan, J. Li, Y. Liu, G. Lu, Y. Xu, Z. Yu, D. Zhang, Focus affinity perception and super-resolution embedding for multifocus image fusion, *IEEE Trans. Neural Netw. Learn. Syst.* (2024).
- [19] Y. Sun, B. Cao, P. Zhu, Q. Hu, Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning, *IEEE Trans. Circuits Syst. Video Technol.* 32 (10) (2022) 6700–6713.
- [20] B. Huang, J. Li, J. Chen, G. Wang, J. Zhao, T. Xu, Anti-uav410: A thermal infrared benchmark and customized scheme for tracking drones in the wild, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [21] S. Razakariyony, F. Jurie, Vehicle detection in aerial imagery: A small target detection benchmark, *J. Vis. Commun. Image Represent.* 34 (2016) 187–203.
- [22] A. Toet, The TNO multiband image data collection, *Data Brief* 15 (2017) 249–251.
- [23] H. Xu, J. Ma, J. Jiang, X. Guo, H. Ling, U2Fusion: A unified unsupervised image fusion network, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (1) (2020) 502–518.
- [24] X. Jia, C. Zhu, M. Li, W. Tang, W. Zhou, LLVIP: A visible-infrared paired dataset for low-light vision, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 3496–3504.
- [25] J. Liu, X. Fan, Z. Huang, G. Wu, R. Liu, W. Zhong, Z. Luo, Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5802–5811.
- [26] L. Tang, J. Yuan, H. Zhang, X. Jiang, J. Ma, PIAFusion: A progressive infrared and visible image fusion network based on illumination aware, *Inf. Fusion* 83 (2022) 79–92.
- [27] Y. Fu, X.-J. Wu, T. Durrani, Image fusion based on generative adversarial network consistent with perception, *Inf. Fusion* 72 (2021) 110–125.
- [28] J. Ma, H. Xu, J. Jiang, X. Mei, X.-P. Zhang, DDcGAN: A dual-discriminator conditional generative adversarial network for multi-resolution image fusion, *IEEE Trans. Image Process.* 29 (2020) 4980–4995.
- [29] J. Gui, Z. Sun, Y. Wen, D. Tao, J. Ye, A review on generative adversarial networks: Algorithms, theory, and applications, *IEEE Trans. Knowl. Data Eng.* 35 (4) (2021) 3313–3332.
- [30] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, *Adv. Neural Inf. Process. Syst.* 33 (2020) 6840–6851.
- [31] Z. Zhao, H. Bai, J. Zhang, Y. Zhang, S. Xu, Z. Lin, R. Timofte, L. Van Gool, Cddfuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 5906–5916.
- [32] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, X. Wang, Vision mamba: Efficient visual representation learning with bidirectional state space model, 2024, *arXiv preprint arXiv:2401.09417*.
- [33] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III* 18, Springer, 2015, pp. 234–241.
- [34] X. Li, F. Zhou, H. Tan, Y. Chen, W. Zuo, Multi-focus image fusion based on nonsubsampling contourlet transform and residual removal, *Signal Process.* 184 (2021) 108062.
- [35] G. Treece, Real image denoising with a locally-adaptive bitonic filter, *IEEE Trans. Image Process.* 31 (2022) 3151–3165.
- [36] K. He, J. Sun, Fast guided filter, 2015, *arXiv preprint arXiv:1505.00996*.
- [37] Y. Jie, F. Zhou, H. Tan, G. Wang, X. Cheng, X. Li, Tri-modal medical image fusion based on adaptive energy choosing scheme and sparse representation, *Measurement* 204 (2022) 112038.
- [38] X. Li, W. Wan, F. Zhou, X. Cheng, Y. Jie, H. Tan, Medical image fusion based on sparse representation and neighbor energy activity, *Biomed. Signal Process. Control.* 80 (2023) 104353.
- [39] X. Li, F. Zhou, H. Tan, Joint image fusion and denoising via three-layer decomposition and sparse representation, *Knowl.-Based Syst.* 224 (2021) 107087.
- [40] Q. Zhang, Y. Liu, R.S. Blum, J. Han, D. Tao, Sparse representation based multi-sensor image fusion for multi-focus and multi-modality images: A review, *Inf. Fusion* 40 (2018) 57–75.
- [41] Y. Liu, X. Chen, A. Liu, R.K. Ward, Z.J. Wang, Recent advances in sparse representation based medical image fusion, *IEEE Instrum. Meas. Mag.* 24 (2) (2021) 45–53.
- [42] B. Aydemir, L. Hoffstetter, T. Zhang, M. Salzmann, S. Süsstrunk, Tempsal-uncovering temporal information for deep saliency prediction, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 6461–6470.
- [43] J. Downes, S. Saltwick, A. Chen, Saliency driven imagery preprocessing for efficient compression-industrial paper, in: *Proceedings of the 31st ACM International Conference on Advances in Geographic Information Systems*, 2023, pp. 1–11.
- [44] Y. Su, J. Deng, R. Sun, G. Lin, H. Su, Q. Wu, A unified transformer framework for group-based segmentation: Co-segmentation, co-saliency detection and video salient object detection, *IEEE Trans. Multimed.* (2023).
- [45] J. Ma, L. Tang, M. Xu, H. Zhang, G. Xiao, StdFusionNet: An infrared and visible image fusion network based on salient target detection, *IEEE Trans. Instrum. Meas.* 70 (2021) 1–13.
- [46] J. Liu, R. Dian, S. Li, H. Liu, Sgfusion: A saliency guided deep-learning framework for pixel-level image fusion, *Inf. Fusion* 91 (2023) 205–214.
- [47] D. Wang, J. Liu, R. Liu, X. Fan, An interactively reinforced paradigm for joint infrared-visible image fusion and saliency object detection, *Inf. Fusion* 98 (2023) 101828.
- [48] J. Liu, X. Fan, J. Jiang, R. Liu, Z. Luo, Learning a deep multi-scale feature ensemble and an edge-attention guidance for image fusion, *IEEE Trans. Circuits Syst. Video Technol.* 32 (1) (2021) 105–119.
- [49] P. Liang, J. Jiang, X. Liu, J. Ma, Fusion from decomposition: A self-supervised decomposition approach for image fusion, in: *European Conference on Computer Vision*, Springer, 2022, pp. 719–735.
- [50] Z. Liu, J. Liu, G. Wu, L. Ma, X. Fan, R. Liu, Bi-level dynamic learning for jointly multi-modality image fusion and beyond, 2023, *arXiv preprint arXiv:2305.06720*.
- [51] B. Cao, Y. Sun, P. Zhu, Q. Hu, Multi-modal gated mixture of local-to-global experts for dynamic image fusion, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 23555–23564.
- [52] Y. Guan, R. Xu, M. Yao, L. Wang, Z. Xiong, Mutual-guided dynamic network for image fusion, in: *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 1779–1788.

- [53] H. Xu, J. Ma, Z. Le, J. Jiang, X. Guo, FusionDn: A unified densely connected network for image fusion, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, 2020, pp. 12484–12491.
- [54] L. Tang, X. Xiang, H. Zhang, M. Gong, J. Ma, Divfusion: Darkness-free infrared and visible image fusion, *Inf. Fusion* 91 (2023) 477–493.
- [55] Z. Huang, J. Liu, X. Fan, R. Liu, W. Zhong, Z. Luo, ReConet: Recurrent correction network for fast and efficient multi-modality image fusion, in: European Conference on Computer Vision, Springer, 2022, pp. 539–555.
- [56] D. Wang, J. Liu, X. Fan, R. Liu, Unsupervised misaligned infrared and visible image fusion via cross-modality image generation and registration, 2022, arXiv preprint arXiv:2205.11876.
- [57] H. Li, J. Liu, Y. Zhang, Y. Liu, A deep learning framework for infrared and visible image fusion without strict registration, *Int. J. Comput. Vis.* 132 (5) (2024) 1625–1644.
- [58] H. Li, J. Zhao, J. Li, Z. Yu, G. Lu, Feature dynamic alignment and refinement for infrared-visible image fusion: Translation robust fusion, *Inf. Fusion* 95 (2023) 26–41.
- [59] H. Zhang, H. Xu, X. Tian, J. Jiang, J. Ma, Image fusion meets deep learning: A survey and perspective, *Inf. Fusion* 76 (2021) 323–336.
- [60] L. Tang, H. Zhang, H. Xu, J. Ma, Rethinking the necessity of image fusion in high-level vision tasks: A practical infrared and visible image fusion network based on progressive semantic injection and scene fidelity, *Inf. Fusion* 99 (2023) 101870.
- [61] Z. Zhao, S. Xu, C. Zhang, J. Liu, P. Li, J. Zhang, DIDFuse: Deep image decomposition for infrared and visible image fusion, 2020, arXiv preprint arXiv:2003.09210.
- [62] H. Li, X.-J. Wu, J. Kittler, RFN-Nest: An end-to-end residual fusion network for infrared and visible images, *Inf. Fusion* 73 (2021) 72–86.
- [63] M. Zhou, J. Huang, N. Zheng, C. Li, Learned image reasoning prior penetrates deep unfolding network for panchromatic and multi-spectral image fusion, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 12398–12407.
- [64] H. Li, T. Xu, X.-J. Wu, J. Lu, J. Kittler, Lrnet: A novel representation learning guided fusion network for infrared and visible images, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [65] Z. Zhao, S. Xu, J. Zhang, C. Liang, C. Zhang, J. Liu, Efficient and model-based infrared and visible image fusion via algorithm unrolling, *IEEE Trans. Circuits Syst. Video Technol.* 32 (3) (2021) 1186–1196.
- [66] H. Zhou, W. Wu, Y. Zhang, J. Ma, H. Ling, Semantic-supervised infrared and visible image fusion via a dual-discriminator generative adversarial network, *IEEE Trans. Multimed.* 25 (2021) 635–648.
- [67] Y. Gao, S. Ma, J. Liu, DCDR-GAN: A densely connected disentangled representation generative adversarial network for infrared and visible image fusion, *IEEE Trans. Circuits Syst. Video Technol.* 33 (2) (2022) 549–561.
- [68] J. Ma, H. Zhang, Z. Shao, P. Liang, H. Xu, GANMcC: A generative adversarial network with multiclassification constraints for infrared and visible image fusion, *IEEE Trans. Instrum. Meas.* 70 (2020) 1–14.
- [69] Y. Rao, D. Wu, M. Han, T. Wang, Y. Yang, T. Lei, C. Zhou, H. Bai, L. Xing, AT-GAN: A generative adversarial network with attention and transition for infrared and visible image fusion, *Inf. Fusion* 92 (2023) 336–349.
- [70] J. Ma, W. Yu, P. Liang, C. Li, J. Jiang, FusionGAN: A generative adversarial network for infrared and visible image fusion, *Inf. Fusion* 48 (2019) 11–26.
- [71] H. Li, X.-J. Wu, DenseFuse: A fusion approach to infrared and visible images, *IEEE Trans. Image Process.* 28 (5) (2018) 2614–2623.
- [72] H. Zhang, H. Xu, Y. Xiao, X. Guo, J. Ma, Rethinking the image fusion: A fast unified image fusion network based on proportional maintenance of gradient and intensity, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, 2020, pp. 12797–12804.
- [73] J. Liu, R. Lin, G. Wu, R. Liu, Z. Luo, X. Fan, CoConet: Coupled contrastive learning network with multi-level feature ensemble for multi-modality image fusion, *Int. J. Comput. Vis.* 132 (5) (2024) 1748–1775.
- [74] X. Yi, H. Xu, H. Zhang, L. Tang, J. Ma, Text-IF: Leveraging semantic text guidance for degradation-aware and interactive image fusion, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 27026–27035.
- [75] J. Zhang, L. Jiao, W. Ma, F. Liu, X. Liu, L. Li, P. Chen, S. Yang, Transformer based conditional GAN for multimodal image fusion, *IEEE Trans. Multimed.* (2023).
- [76] Y. Sun, B. Cao, P. Zhu, Q. Hu, Detfusion: A detection-driven infrared and visible image fusion network, in: Proceedings of the 30th ACM International Conference on Multimedia, 2022, pp. 4003–4011.
- [77] L. Tang, J. Yuan, J. Ma, Image fusion in the loop of high-level vision tasks: A semantic-aware real-time infrared and visible image fusion network, *Inf. Fusion* 82 (2022) 28–42.
- [78] J. Liu, Z. Liu, G. Wu, L. Ma, R. Liu, W. Zhong, Z. Luo, X. Fan, Multi-interactive feature learning and a full-time multi-modality benchmark for image fusion and segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 8115–8124.
- [79] J. Liu, G. Wu, Z. Liu, L. Ma, R. Liu, X. Fan, Where elegance meets precision: Towards a compact, automatic, and flexible framework for multi-modality image fusion and applications, in: *IJCAI*, 2024.
- [80] H. Zhang, X. Zuo, J. Jiang, C. Guo, J. Ma, MRFS: Mutually reinforcing image fusion and segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 26974–26983.
- [81] X. Deng, P.L. Dragotti, Deep convolutional neural network for multi-modal image restoration and fusion, *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (10) (2020) 3333–3348.
- [82] J. Li, D. Li, C. Xiong, S. Hoi, Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation, in: International Conference on Machine Learning, PMLR, 2022, pp. 12888–12900.
- [83] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PMLR, 2021, pp. 8748–8763.
- [84] Z. Luo, F.K. Gustafsson, Z. Zhao, J. Sjölund, T.B. Schön, Controlling vision-language models for universal image restoration, 2023, arXiv preprint arXiv:2310.01018.
- [85] C. Yang, Z. An, L. Huang, J. Bi, X. Yu, H. Yang, B. Diao, Y. Xu, CLIP-KD: An empirical study of CLIP model distillation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 15952–15962.
- [86] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 10684–10695.
- [87] H. Chung, B. Sim, J.C. Ye, Come-closer-diffuse-faster: Accelerating conditional diffusion models for inverse problems through stochastic contraction, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 12413–12422.
- [88] O. Avrahami, D. Lischinski, O. Fried, Blended diffusion for text-driven editing of natural images, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 18208–18218.
- [89] J. Choi, S. Kim, Y. Jeong, Y. Gwon, S. Yoon, Ilvr: Conditioning method for denoising diffusion probabilistic models, 2021, arXiv preprint arXiv:2108.02938.
- [90] A. Lugmayr, M. Danelljan, A. Romero, F. Yu, R. Timofte, L. Van Gool, Repaint: Inpainting using denoising diffusion probabilistic models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11461–11471.
- [91] A. Rahman, J.M.J. Valanarasu, I. Hacıhaliloglu, V.M. Patel, Ambiguous medical image segmentation using diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 11536–11546.
- [92] J. Xu, S. Liu, A. Vahdat, W. Byeon, X. Wang, S. De Mello, Open-vocabulary panoptic segmentation with text-to-image diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 2955–2966.
- [93] R.S. Zimmermann, L. Schott, Y. Song, B.A. Dunn, D.A. Klindt, Score-based generative classifiers, 2021, arXiv e-prints.
- [94] F.-A. Croitoru, V. Hondru, R.T. Ionescu, M. Shah, Diffusion models in vision: A survey, *IEEE Trans. Pattern Anal. Mach. Intell.* (2023).
- [95] C. Saharia, J. Ho, W. Chan, T. Salimans, D.J. Fleet, M. Norouzi, Image super-resolution via iterative refinement, *IEEE Trans. Pattern Anal. Mach. Intell.* 45 (4) (2022) 4713–4726.
- [96] A. Hatamizadeh, J. Song, G. Liu, J. Kautz, A. Vahdat, Diffit: Diffusion vision transformers for image generation, in: European Conference on Computer Vision, Springer, 2025, pp. 37–55.
- [97] J. Yue, L. Fang, S. Xia, Y. Deng, J. Ma, Dif-fusion: Towards high color fidelity in infrared and visible image fusion with diffusion models, *IEEE Trans. Image Process.* (2023).
- [98] B. Yang, Z. Jiang, D. Pan, H. Yu, G. Gui, W. Gui, LFDT-Fusion: A latent feature-guided diffusion transformer model for general image fusion, *Inf. Fusion* 113 (2025) 102639.
- [99] X. Yi, L. Tang, H. Zhang, H. Xu, J. Ma, Diff-IF: Multi-modality image fusion via diffusion model with fusion knowledge prior, *Inf. Fusion* 110 (2024) 102450.
- [100] L. Tang, Y. Deng, X. Yi, Q. Yan, Y. Yuan, J. Ma, DRMF: Degradation-robust multi-modal image fusion via composable diffusion prior, in: Proceedings of the 32nd ACM International Conference on Multimedia, 2024, pp. 8546–8555.
- [101] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, 2023, arXiv preprint arXiv:2312.00752.
- [102] J. Liu, G. Wu, Z. Liu, D. Wang, Z. Jiang, L. Ma, W. Zhong, X. Fan, Infrared and visible image fusion: From data compatibility to task adaption, *IEEE Trans. Pattern Anal. Mach. Intell.* (2024).
- [103] K. He, X. Zhang, S. Ren, J. Sun, Delving deep into rectifiers: Surpassing human-level performance on imagenet classification, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1026–1034.
- [104] K. Song, X. Xue, H. Wen, Y. Ji, Y. Yan, Q. Meng, Misaligned visible-thermal object detection: A drone-based benchmark and baseline, *IEEE Trans. Intell. Veh.* (2024).
- [105] D. Summers, Harvard whole brain atlas: www.med.harvard.edu/AAN-LIB/home.html, *J. Neurol. Neurosurg. Psychiatry* 74 (3) (2003) 288.

- [106] R. Wu, T. Yang, L. Sun, Z. Zhang, S. Li, L. Zhang, Seesr: Towards semantics-aware real-world image super-resolution, 2023, arXiv preprint arXiv:2311.16518.
- [107] J. Wang, Z. Yue, S. Zhou, K.C. Chan, C.C. Loy, Exploiting diffusion prior for real-world image super-resolution, 2023, arXiv preprint arXiv:2305.07015.
- [108] Q. Ming, Z. Zhou, L. Miao, H. Zhang, L. Li, Dynamic anchor learning for arbitrary-oriented object detection, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, 2021, pp. 2355–2363.
- [109] J. Liu, X. Li, Z. Wang, Z. Jiang, W. Zhong, W. Fan, B. Xu, PromptFusion: Harmonized semantic prompt learning for infrared and visible image fusion, IEEE/CAA J. Autom. Sin. (2024).
- [110] J. Li, J. Chen, J. Liu, H. Ma, Learning a graph neural network with cross modality interaction for image fusion, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 4471–4479.
- [111] D. Rao, T. Xu, X.-J. Wu, TGFuse: An infrared and visible image fusion approach based on transformer and generative adversarial network, IEEE Trans. Image Process. (2023).
- [112] Y. Han, Y. Cai, Y. Cao, X. Xu, A new image fusion performance metric based on visual information fidelity, Inf. Fusion 14 (2) (2013) 127–135.
- [113] C.S. Xydeas, V. Petrovic, et al., Objective image fusion performance measure, Electron. Lett. 36 (4) (2000) 308–309.
- [114] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE Trans. Image Process. 13 (4) (2004) 600–612.
- [115] R. Zhang, P. Isola, A.A. Efros, E. Shechtman, O. Wang, The unreasonable effectiveness of deep features as a perceptual metric, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 586–595.
- [116] A.Q. Nichol, P. Dhariwal, Improved denoising diffusion probabilistic models, in: International Conference on Machine Learning, PMLR, 2021, pp. 8162–8171.
- [117] X. Li, J. Liu, Z. Chen, Y. Zou, L. Ma, X. Fan, R. Liu, Contourlet residual for prompt learning enhanced infrared image super-resolution, in: European Conference on Computer Vision, Springer, 2024, pp. 270–288.
- [118] G. Jocher, A. Chaurasia, J. Qiu, Ultralytics YOLO, 2023, URL <https://github.com/ultralytics/ultralytics>.
- [119] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, N. Sang, Bisenet: Bilateral segmentation network for real-time semantic segmentation, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 325–341.
- [120] M. Li, R. Pei, T. Zheng, Y. Zhang, W. Fu, FusionDiff: Multi-focus image fusion using denoising diffusion probabilistic models, Expert Syst. Appl. 238 (2024) 121664.
- [121] C. Cheng, T. Xu, X.-J. Wu, MUFusion: A general unsupervised image fusion network based on memory unit, Inf. Fusion 92 (2023) 80–92.
- [122] M. Nejati, S. Samavi, S. Shirani, Multi-focus image fusion using dictionary-based sparse representation, Inf. Fusion 25 (2015) 72–84.
- [123] H. Zhang, Z. Le, Z. Shao, H. Xu, J. Ma, MFF-GAN: An unsupervised generative adversarial network with adaptive and gradient joint constraints for multi-focus image fusion, Inf. Fusion 66 (2021) 40–53.